Aircraft landing*

Mariana Cristino[1[up202502528]], and Patrícia Silva[1[up202202895]]

[1] Faculty of Engineering of the University of Porto, Rua Dr. Roberto Frias, s/n 4200-465 Porto, Portugal.

Abstract. This paper studies the Aircraft Landing Problem (ALP), focusing on the scheduling of landing times and runway assignments under time window and safety separation constraints. Three modeling approaches are compared: Mixed-Integer Programming (MIP), Constraint Programming (CP), and a Hybrid Logic-Based Benders Decomposition (LBBD). Formulations for both single and multi-runway settings are presented. Computational results on benchmark instances show that MIP performs well on small problems but faces scalability limits beyond 100 aircraft, while CP and Hybrid approaches remain effective on larger, multi-runway instances. The impact of search and decision strategies is also analyzed, highlighting trade-offs between exact optimization and scalable constraint-based methods.

Keywords: Air traffic control · Scheduling Optimization

1 Introduction

The Aircraft Landing Problem (ALP) constitutes a fundamental challenge in air traffic management, particularly as global air traffic density continues to escalate. At its core, the ALP involves determining precise landing times for a designated set of aircraft on one or more available runways. The primary objective is to optimize the landing schedule while strictly adhering to critical operational constraints.

This report presents a comprehensive investigation into various computational models designed to solve the static ALP, where all information regarding the aircraft set is known a priori.

The principal objective of this paper is to evaluate the scalability and efficiency of different modeling paradigms: Mixed-Integer Programming (MIP), Constraint Programming (CP), and a Hybrid Logic-Based Benders Decomposition. Furthermore, this work analyzes the impact of diverse search and decision strategies within CP and provides a comparative analysis of MIP and CP.

2 Problem Context

2.1 Fundamental Problem Description

The Aircraft Landing Problem arises when an aircraft enters the radar horizon of an airport, and Air Traffic Control (ATC) must assign it a precise landing time and, in multi-runway settings, an appropriate runway.

Each aircraft is associated with a feasible landing window, bounded by an earliest and a latest landing time. The earliest time corresponds to flying at maximum feasible speed, while the latest reflects fuel-efficient operation combined with limited holding capacity. In addition, every aircraft has a target landing time, achieved under optimal cruise conditions. Deviations from this target incur penalty costs, which increase with the magnitude of the temporal displacement, whether the aircraft lands early or late.

Operational safety further constrains the problem through mandatory separation times between consecutive landings. These separations are required to mitigate wake turbulence effects, which depend on aircraft characteristics, with heavier aircraft typically requiring larger buffers to ensure safe operations.

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2.2 Practical Considerations and Modeling Assumptions

Real-world aircraft landing scheduling involves complexities beyond the basic ALP formulation. The models considered in this study incorporate several assumptions to balance realism with computational tractability.

First, the models function as decision-support tools rather than detailed control systems. They focus on determining feasible and cost-efficient schedules, assuming that ATC can execute the necessary maneuvers to implement any feasible solution. This abstraction enables strategic analysis, such as identifying congestion patterns or evaluating runway capacity.

Second, separation times are modeled at a granular level, allowing aircraft-pair-specific separations instead of broad aircraft classes. This better reflects heterogeneous fleets and operational constraints.

Third, finite latest landing times are explicitly considered. Unlike formulations that allow unrealistically large time horizons, this approach captures fuel and holding limitations and enables the detection of infeasible schedules under tight conditions.

The models also address runway allocation decisions in multi-runway environments, supporting both segregated and mixed-mode operations. This allows different runway usage policies to be compared within a unified framework.

Finally, total deviation costs are minimized using linear or piecewise-linear objective functions. While simplified relative to real operational costs, this choice ensures compatibility with Mixed-Integer Programming and Constraint Programming approaches. The framework remains flexible enough to reflect different stakeholder perspectives, such as minimizing congestion for airport operators or reducing delays and costs for airlines.

3 Mixed-Integer Programming (MIP) Model

3.1 Single Runway Formulation

In this section, we present the mixed-integer zero-one formulation for the static single runway aircraft landing problem. This model aims to determine the landing time for each aircraft while minimizing deviation costs and respecting safety separations.

Notation

Indices and Sets:
- i, j: indices for planes where $i, j \in \{1, ..., P\}$.
- U: set of pairs (i, j) with uncertain landing order (overlapping time windows).
- V: set of pairs (i, j) where i definitely lands before j, but separation is not guaranteed.
- W: set of pairs (i, j) where i definitely lands before j and separation is naturally satisfied.

Parameters:
- P: Total number of planes.
- E_i: Earliest allowable landing time for plane i.
- L_i: Latest allowable landing time for plane i.
- T_i: Target landing time for plane i.
- $S_{i,j}$: Required separation time between plane i and plane j, provided i lands before j.
- g_i: Penalty costs per unit of time for landing before T_i.
- h_i: Penalty costs per unit of time for landing after T_i.

Decision Variables:
- x_i: Scheduled landing time for plane i.
- α_i: Earliness deviation ($\alpha_i \geq 0$).
- β_i: Lateness deviation ($\beta_i \geq 0$).
- $\delta_{i,j}$: Binary variable, equal to 1 if plane i lands before plane j, and 0 otherwise.

Objective Function

The objective is to minimise the total penalty cost associated with deviations from the target landing times:

$$\min Z = \sum_{i=1}^{P} (g_i \alpha_i + h_i \beta_i) \tag{1}$$

Constraints

1. Time-Window Constraints: Each aircraft must land within its specified window $[E_i, L_i]$.
$$E_i \leq x_i \leq L_i, \quad \forall i \in \{1, ..., P\} \tag{2}$$
2. Landing Order Constraints: For every pair (i, j) where the order is not predetermined, either i precedes j or vice versa.
$$\delta_{i,j} + \delta_{j,i} = 1, \quad \forall (i, j) \in U, i < j \tag{3}$$
3. Predefined Sequence Constraints:
$$\delta_{i,j} = 1, \quad \forall (i, j) \in W \cup V \tag{4}$$
4. Separation Constraints: For pairs in V where the order is known but separation is not guaranteed:
$$x_j \geq x_i + S_{i,j}, \quad \forall (i, j) \in V \tag{5}$$

For pairs in U where the order must be decided:
$$x_j \geq x_i + S_{i,j}\delta_{i,j} - \left(L_i - E_j\right)\delta_{j,i}, \quad \forall (i, j) \in U \tag{6}$$

5. Deviation and Linearity Constraints: These constraints define α_i and β_i while ensuring a linear formulation:
$$x_i = T_i - \alpha_i + \beta_i, \quad \forall i \in \{1, ..., P\} \tag{7}$$
$$\alpha_i \geq T_i - x_i, \quad 0 \leq \alpha_i \leq T_i - E_i \tag{8}$$
$$\beta_i \geq x_i - T_i, \quad 0 \leq \beta_i \leq L_i - T_i \tag{9}$$
6. Variable Domains: Permissible values for all decision variables:
$$\alpha_i, \beta_i, x_i \geq 0, \quad \delta_{i,j} \in \{0, 1\} \tag{10}$$

Sets Definition

The sets are constructed based on the temporal relationship between aircraft time windows:

- $W = \left\{(i, j) \mid L_i < E_j \wedge L_i + S_{i,j} \leq E_j; i \neq j, i = 1, ..., P, j = 1, ..., P\right\}$
- $V = \left\{(i, j) \mid L_i < E_j \wedge L_i + S_{i,j} > E_j; i \neq j, i = 1, ..., P, j = 1, ..., P\right\}$
- $U = \left\{(i, j) \mid E_j \leq E_i \leq L_j \vee E_j \leq L_i \leq L_j \vee E_i \leq E_j \leq L_i \vee E_i \leq L_j \leq L_i; i \neq j, i = 1, ..., P, j = 1, ..., P\right\}$.

Logic Deduction for Equation 6

For pairs in U, a "Big-M" formulation was used to enforce separation only when a specific landing order is chosen. The general form is:
$$x_j \geq x_i + S_{i,j} - M\delta_{j,i} \tag{11}$$

1. If $\delta_{i,j} = 1$ (plane i before j), then $\delta_{j,i} = 0$. The constraint becomes $x_j \geq x_i + S_{i,j}$, enforcing the required separation.
2. If $\delta_{i,j} = 0$ (plane j before i), then $\delta_{j,i} = 1$. The constraint becomes $x_j \geq x_i + S_{i,j} - M$. By setting M to a sufficiently large value, the constraint becomes redundant (inactive).

To improve the LP relaxation, we set $M = L_i + S_{i,j} - E_j$. Substituting M and using the identity $\delta_{i,j} + \delta_{j,i} = 1$:
$$x_j \geq x_i + S_{i,j}\left(\delta_{i,j} + \delta_{j,i}\right) - \left(L_i + S_{i,j} - E_j\right)\delta_{j,i} \tag{12}$$
$$x_j \geq x_i + S_{i,j}\delta_{i,j} + S_{i,j}\delta_{j,i} - L_i\delta_{j,i} - S_{i,j}\delta_{j,i} + E_j\delta_{j,i} \tag{13}$$
$$x_j \geq x_i + S_{i,j}\delta_{i,j} - \left(L_i - E_j\right)\delta_{j,i} \tag{14}$$

Checking the validity for $\delta_{j,i} = 1$:
$$x_j \geq x_i - L_i + E_j \Rightarrow x_j \geq E_j + (x_i - L_i) \tag{15}$$

Since $(x_i - L_i) \leq 0$, this simply implies $x_j \geq E_j + [\text{value} \leq 0]$, which is always satisfied as $x_j \geq E_j$ by definition.

3.2 Multiple Runway Formulation

As air traffic density increases, airports operate with multiple runways to enhance capacity. This necessitates assigning each aircraft to a specific runway.

Notation

Indices:
- r: index for runways where $r \in \{1, ..., R\}$.

Parameters:
- R: Total number of runways.
- $S_{i,j}$: Required separation time between plane i and plane j on different runways.

Decision Variables:
- $y_{i,r}$: 1 if plane i lands on runway r, 0 otherwise.
- $z_{i,j}$: 1 if planes i and j land on the same runway, 0 otherwise.

New Constraints

7. Runway Assignment: Each aircraft must be assigned to exactly one runway.
$$\sum_{r=1}^{R} y_{i,r} = 1, \quad \forall i \in \{1, ..., P\} \tag{16}$$
8. Symmetry Constraint: If i and j share a runway, then j and i also do.
$$z_{i,j} = z_{j,i}, \quad \forall i < j \tag{17}$$
9. Same Runway Logic: Forces $z_{i,j}$ to 1 if both planes share any runway r.
$$z_{i,j} \geq y_{i,r} + y_{j,r} - 1, \quad \forall i < j, r \in \{1, ..., R\} \tag{18}$$

Revised Separation Constraints

The separation now depends on whether aircraft share a runway ($S_{i,j}$) or land on different ones ($s_{i,j}$).

10. Revised Separation for V:
$$x_j \geq x_i + S_{i,j} z_{i,j} + s_{i,j}(1 - z_{i,j}), \quad \forall (i,j) \in V \tag{19}$$
11. Revised Separation for U:

Let $m_s = \max(S_{i,j}, s_{i,j})$:
$$x_j \geq x_i + S_{i,j} z_{i,j} + s_{i,j}(1 - z_{i,j}) - (L_i + m_s - E_j)\delta_{j,i}, \quad \forall (i,j) \in U \tag{20}$$

Logic Deduction for Equation 20

The general Big-M form for multiple runways is:
$$x_j \geq x_i + \left[S_{i,j} z_{i,j} + s_{i,j}(1 - z_{i,j})\right] - M\delta_{j,i} \tag{21}$$
The term $\left[S_{i,j} z_{i,j} + s_{i,j}(1 - z_{i,j})\right]$ ensures the correct buffer: $S_{i,j}$ if $z_{i,j} = 1$ and $S_{i,j}$ if $z_{i,j} = 0$.

To strengthen the LP relaxation, we replace M with $\left(L_i + \max(S_{i,j}, S_{i,j}) - E_j\right)$.

Let $\delta_{i,j} = S_{i,j} z_{i,j} + S_{i,j}(1 - z_{i,j})$ and $\hat{S}_{\{ij\}} = \max(S_{i,j}, S_{i,j})$. The constraint is:
$$x_j \geq x_i + \delta_{i,j} - \left(L_i + \hat{S}_{\{ij\}} - E_j\right)\delta_{j,i} \tag{22}$$

1. If $\delta_{i,j} = 1$, then $\delta_{j,i} = 0$. The constraint simplifies to $x_j \geq x_i + \delta_{i,j}$, enforcing the correct separation.
2. If $\delta_{i,j} = 0$, then $\delta_{j,i} = 1$. The constraint becomes:

$x_j \geq E_j + (x_i - L_i) + \left(\delta_{i,j} - \hat{S}_{\{ij\}}\right)$ Since $(x_i - L_i) \leq 0$ and $\delta_{i,j} \leq \hat{S}_{\{ij\}}$, the right-hand side is always $\leq E_j$, making the constraint redundant.

4 Constraint Programming (CP) Model

4.1 Single Runway

In this section, we present the Constraint Programming formulation for the static single runway aircraft landing problem. Unlike the MIP approach, CP allows for the direct use of logical implications and non-linear operators.

Notation

Indices and Sets:

- i, j: indices for planes where $i, j \in \{1, ..., P\}$.
- U: set of pairs (i, j) with uncertain landing order.
- V: set of pairs (i, j) where i definitely lands before j, but separation is not guaranteed.
- W: set of pairs (i, j) where i definitely lands before j and separation is naturally satisfied.

Parameters:

- P: Total number of planes.
- E_i, L_i: Earliest and latest allowable landing times for plane i.
- T_i: Target landing time for plane i.
- $S_{i,j}$: Required separation time between plane i and plane j, provided i lands before j.
- g_i, h_i: Penalty costs per unit of time for earliness and lateness.

Decision Variables:

- x_i: Scheduled landing time for plane i, where $x_i \in [E_i, L_i]$.
- α_i: Earliness deviation ($\alpha_i \geq 0$).
- β_i: Lateness deviation ($\beta_i \geq 0$).
- $b_{i,j}$: Binary variable, equal to 1 if plane i lands before plane j, and 0 otherwise.

Objective Function

The objective remains the minimization of total penalty costs:

$$\min Z = \sum_{i=1}^{P} (g_i \alpha_i + h_i \beta_i) \tag{23}$$

Constraints

1. Time-Window Constraints:
$$E_i \leq x_i \leq L_i, \quad \forall i \in \{1, ..., P\} \tag{24}$$
2. Order Uniqueness: For every pair (i, j) where the order is not predetermined, either i precedes j or vice versa.
$$b_{i,j} + b_{j,i} = 1, \quad \forall (i, j) \in U, i < j \tag{25}$$
3. Deviation Definitions:
$$\alpha_i = \max(0, T_i - x_i), \quad \forall i \in \{1, ..., P\} \tag{26}$$
$$\beta_i = \max(0, x_i - T_i), \quad \forall i \in \{1, ..., P\} \tag{27}$$
4. Fixed Sequence Separation: For pairs in V (where i always precedes j):
$$x_j \geq x_i + S_{i,j}, \quad \forall (i, j) \in V \tag{28}$$
5. Conditional Separation: For pairs in U, separation is enforced based on the decided order:

$b_{i,j} = 1 \Rightarrow x_j \geq x_i + S_{i,j}, \quad \forall (i, j) \in U$
$$b_{j,i} = 1 \Rightarrow x_i \geq x_j + S_{j,i}, \quad \forall (i, j) \in U \tag{29}$$

Sets Definition

- $W = \{(i, j) \mid L_i < E_j \land L_i + S_{i,j} \leq E_j; i \neq j\}$
- $V = \{(i, j) \mid L_i < E_j \land L_i + S_{i,j} > E_j; i \neq j\}$
- $U = \{(i, j) \mid E_j \leq E_i \leq L_j \lor E_j \leq L_i \leq L_j \lor E_i \leq E_j \leq L_i \lor E_i \leq L_j \leq L_i; i \neq j\}$

4.2 Multiple Runways

The CP approach simplifies the transition to multiple runways by using a discrete decision variable for runway assignment and logical connectors for conditional constraints.

Notation

Parameters:

- R: Total number of runways.
- $s_{i,j}$: Required separation time between plane i and plane j when landing on different runways.

Decision Variables:

- r_i: Index of the runway assigned to plane i, where $r_i \in \{1, ..., R\}$.

New and Revised Constraints

6. Sequence Separation for V: For planes with a fixed order, the buffer depends on the runway assignment:

$$(r_i = r_j) \Rightarrow x_j \geq x_i + S_{i,j}, \quad \forall (i,j) \in V \tag{30}$$

$$(r_i \neq r_j) \Rightarrow x_j \geq x_i + s_{i,j}, \quad \forall (i,j) \in V \tag{31}$$

7. Conditional Separation for U:

$$(b_{i,j} = 1 \land r_i = r_j) \Rightarrow x_j \geq x_i + S_{i,j}, \quad \forall (i,j) \in U \tag{32}$$

$$(b_{i,j} = 1 \land r_i \neq r_j) \Rightarrow x_j \geq x_i + s_{i,j}, \quad \forall (i,j) \in U \tag{33}$$

$$(b_{j,i} = 1 \land r_i = r_j) \Rightarrow x_i \geq x_j + S_{j,i}, \quad \forall (i,j) \in U \tag{34}$$

$$(b_{j,i} = 1 \land r_i \neq r_j) \Rightarrow x_i \geq x_j + s_{j,i}, \quad \forall (i,j) \in U \tag{35}$$

5 Hybrid Logic-Based Benders Decomposition (LBBD)

This approach employs a CP-LP decomposition strategy where a Constraint Programming Master Problem handles discrete decisions (sequencing and allocation), while a Linear Programming Subproblem handles continuous scheduling. A distinguishing feature of this formulation is the use of a "Strengthened Method", which includes proxy time variables to enforce coarse temporal feasibility and improve the lower bound estimate before invoking the subproblem.

Notation

Indices and Sets:

- i, j: Indices for planes in $P = \{1, ..., P\}$.
- U: Set of pairs (i, j) with uncertain landing order.
- V: Set of pairs (i, j) with fixed order but uncertain separation.
- R: Set of available runways $\{1, ..., R\}$.

Parameters:

- g_i, h_i: Penalty costs for earliness and lateness.
- $S_{i,j}, s_{i,j}$: Separation times for same and different runways, respectively.

Decision Variables:

- r_i: Integer variable for the runway assigned to plane i ($r_i \in R$).
- $b_{i,j}$: Boolean variable, equal to 1 if i precedes j for $(i, j) \in U$.
- θ: Integer variable representing the estimated total cost (Benders estimator).

Proxy Variables (Strengthening):

- x_i^m: Proxy integer landing time for plane i ($x_i^m \in [E_i, L_i]$).
- α_i^m, β_i^m: Proxy integer earliness and lateness deviations.

5.1 Master Problem

The Master Problem determines the runway assignment and the landing sequence for uncertain pairs, minimizing an estimator variable θ.

Objective Function

The Master minimizes the estimated cost θ, which is bounded by the internal proxy cost calculation:

$$\min \theta \tag{36}$$

Constraints

1. Order Exclusivity: For uncertain pairs, a strict linear order must be established.

$$b_{i,j} + b_{j,i} = 1, \quad \forall (i,j) \in U \tag{37}$$

2. Proxy Deviation Definition: Relates proxy landing times to target times to estimate costs.

$$x_i^m + \alpha_i^m - \beta_i^m = T_i, \quad \forall i \in P \tag{38}$$

3. Logical Separation Constraints: These constraints enforce temporal feasibility within the Master using logical implication. Let $\Delta_{i,j}$ be the effective separation based on runway assignment:

$$\Delta_{i,j} = \begin{cases} S_{i,j} & \text{if } r_i = r_j \\ s_{i,j} & \text{if } r_i \neq r_j \end{cases} \tag{39}$$

For pairs in V (fixed order):

$$x_j^m \geq x_i^m + \Delta_{i,j}, \quad \forall (i,j) \in V \tag{40}$$

For pairs in U (uncertain order), separation is enforced only if i precedes j:

$$b_{i,j} = 1 \Rightarrow x_j^m \geq x_i^m + \Delta_{i,j}, \quad \forall (i,j) \in U \tag{41}$$

4. Lower Bound Constraint: The estimator θ must be at least the cost calculated by the proxy variables.

$$\theta \geq \sum_{i=1}^{P} (g_i \alpha_i^m + h_i \beta_i^m) \tag{42}$$

5.2 Subproblem (LP)

The subproblem is a continuous LP solved for a fixed runway assignment $\overline{r}$ and sequence $\overline{b}$ provided by the Master.

Variables: x_i, α_i, β_i (Continuous).

Formulation

$$\min Z_{\text{SP}} = \sum_{i=1}^{P} (g_i \alpha_i + h_i \beta_i) \tag{43}$$

Subject to:
1. Exact Deviation Definition:

$$x_i + \alpha_i - \beta_i = T_i, \quad \forall i \in P \tag{44}$$

2. Separation constraints induced by the fixed order:
For every pair (i,j) for which the subproblem considers “i precedes j”, define the effective separation:

$$\Delta_{i,j} = \begin{cases} S_{i,j} & \text{if } \overline{r}_i = \overline{r}_j \\ s_{i,j} & \text{if } \overline{r}_i \neq \overline{r}_j \end{cases} \tag{45}$$

Then impose:

$$x_j \geq x_i + \Delta_{i,j}(\overline{r}), \quad \forall (i,j) \in P\left(\overline{b}\right) \tag{46}$$

where the enforced precedence set is:

$$P\left(\overline{b}\right) = V \cup W \cup \left\{(i,j) \in U : \overline{b}_{ij} = 1\right\} \tag{47}$$

where $\Delta_{i,j}(\overline{r})$ is the separation constant determined by whether $\overline{r}_i = \overline{r}_j$.

5.3 Benders Cuts

The Master and Subproblem communicate via cuts added to the Master problem in subsequent iterations. Let k denote the current iteration with fixed solution $\left(\overline{r}^k, \overline{b}^k\right)$.

Infeasibility Cuts (No-Good)

If the subproblem is infeasible for $\left(\overline{r}, \overline{b}\right)$, the specific assignment configuration is forbidden:

$$\neg\left(\bigwedge_{i \in P} (r_i = \overline{r}_i) \wedge \bigwedge_{(i,j) \in U} \left(b_{i,j} = \overline{b}_{i,j}\right) \tag{48}$$

Optimality Cuts

If the Master proposes $\left(\overline{r}, \overline{b}\right)$ and the LP returns cost $z\left(\overline{r}, \overline{b}\right)$, then whenever the Master chooses the same combination again, θ must be at least that cost:

$$\left(\bigwedge_{i \in P} (r_i = \overline{r}_i) \wedge \bigwedge_{(i,j) \in U} \left(b_{i,j} = \overline{b}_{i,j}\right) \right) \Rightarrow \theta \geq \left\lceil z\left(\overline{r}, \overline{b}\right) \right\rceil \tag{49}$$

This LBBD formulation iteratively refines the master problem with cuts until convergence.

6 Decision and Search Strategies

This section describes the decision, branching, and search strategies used in the CP, MIP, and Hybrid approaches. Whether explicitly defined or handled by solver heuristics, these strategies have a strong influence on execution time and schedule quality.

6.1 MIP Search Strategies

In the MIP formulation, the optimization process is primarily driven by the internal branch-and-bound mechanism of the `SCIP` solver accessed through Google OR-Tools. Variable selection, branching decisions, and node exploration are handled automatically using solver-specific heuristics such as LP relaxations.

The main intervention is hinting: initial values for landing times are set to each aircraft's target time using `SetHint`. Since optimal schedules in the problem often lie close to target times, this approach leads the solver to high-quality regions of the search space, reducing exploration of suboptimal branches and accelerating convergence without altering the model.

6.2 CP and Hybrid Search Strategies

The CP models and the Master Problem of the Hybrid LBBD approach use the `CP-SAT` solver, allowing explicit control over the search process through branching and decision strategies.

Search Branching Policies

The CP solver provides several global search strategies:

- Automatic Search (`AUTOMATIC_SEARCH`): Solver chooses heuristics dynamically, serves as a baseline.
- Fixed Search (`FIXED_SEARCH`): Follows the order in which variables are declared, e.g., sequentially by aircraft index. Useful when indices reflect temporal order.
- Portfolio Search (`PORTFOLIO_SEARCH`): Alternates between multiple heuristics to balance exploration and exploitation.
- LP Search (`LP_SEARCH`): Guides branching using LP relaxation values, prioritizing promising assignments, effective for time-based problems like ALP.

These strategies influence tree exploration and affect runtime and scalability.

Variable and Value Selection

Decision strategies can further specify:

- Variable selection rules: Which variable is instantiated next (landing times, runway assignments).
- Value selection rules: Which candidate values are tested first.

This allows fine-grained control of the search and experimentation with different behaviors.

Use of Hints

Hints suggest landing times near target times, guiding the solver toward realistic, low-cost solutions and reducing exploration of poor-quality regions. This improves efficiency and accelerates convergence, similarly to the MIP model.

6.3 Hybrid Approach (CP–LP with LBBD)

In the Hybrid LBBD framework:

- Master Problem (CP): Handles combinatorial decisions (runways, sequencing, proxy costs). CP search strategies guide the solver to high-quality candidate sequences.
- Subproblem (LP): Computes optimal landing times given fixed sequences. Solved deterministically, without branching.

Iterations between master and subproblem add feasibility or optimality cuts as needed, combining CP for combinatorial structure with LP for continuous optimization. This separation leverages CP for combinatorial choices and LP for continuous optimization.

Decision and search strategies significantly impact performance. MIP relies mainly on solver heuristics with hinting; CP enables explicit search control; Hybrid combines CP for combinatorial decisions with LP for continuous optimization. These approaches provide a scalable and efficient framework for the Aircraft Landing Problem across various instance sizes.

7 Experimental Setup and Dataset Configuration

From Dataset 1 to Dataset 13, all models (MIP, CP, and Hybrid) were systematically tested across all instances. For single-runway configurations, instances from Dataset 9 onward were excluded due to computational infeasibility, as these cases combine a high number of aircraft with wide freeze-time windows, leading to rapid growth of the branch-and-bound search space and prohibitive runtimes.

For multiple-runway scenarios, Datasets 1 to 8 were solved by iteratively increasing the number of runways until an optimal solution with zero total penalty was reached. This approach ensured that the solver identified the minimal number of runways required to fully eliminate deviations, effectively performing a search for the best runway allocation per instance. For Datasets 9 to 13, only multi-runway configurations were considered, as single-runway optimization was impractical.

These experimental choices provide a comprehensive evaluation of all models while respecting computational limits. Additional details on complexity estimates and the specific runway configurations used are provided in the Appendix (Tables 1 and 2).

8 Visualization

To support the analysis and interpretation of aircraft landing schedules, detailed visualizations were generated. Fig. 1 presents an example of a solution visualization. The figure shows, for each aircraft, its available landing window through horizontal lines representing the earliest and latest permissible landing times, together with the target landing time and the scheduled landing time assigned by the solver. Deviations from the target are explicitly highlighted, with distinct visual indicators for early and late landings, while aircraft landing exactly at their target times are marked with a special symbol.

In multi-runway scenarios, aircraft are grouped by runway and ordered according to their landing times, improving readability and allowing an immediate assessment of runway utilization. In addition, key solution metrics are displayed alongside the schedule, providing contextual information on deviations, penalties, and overall solver performance.

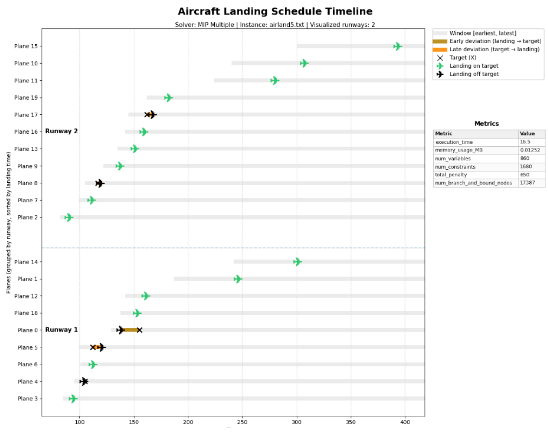

Fig. 1. Solution with the MIP in the multiple runway scenario

Overall, this visualization consolidates scheduling constraints, solver decisions, and penalty information into a single figure, facilitating interpretation, comparison between solutions, and qualitative evaluation of solution quality.

9 Performance Metrics

To evaluate and compare the performance of the MIP, CP, and Hybrid models, a common set of performance metrics was collected across all approaches. These metrics are designed to capture both computational efficiency and solver behavior, enabling a systematic analysis of scalability, robustness, and search effectiveness.

Across all models, two baseline metrics were consistently measured:

- **Total execution time**, capturing the end-to-end computational effort.
- **Peak memory usage (MB)**, reflecting the resource footprint of each approach.

These two metrics provide a first-order comparison of efficiency and practicality, particularly relevant when assessing large-scale instances.

Beyond these common indicators, model-specific metrics were selected to reflect the internal mechanisms of each solving paradigm.

9.1 MIP Performance Metrics

For the MIP model, performance evaluation focuses on indicators related to model size and branch-and-bound complexity, which are the dominant drivers of computational effort in exact optimization.

- **Number of variables** and **constraints**, measuring model size and structural complexity. These values grow rapidly with the number of aircraft and directly impact LP relaxation strength and memory usage.
- **Total penalty (objective value)**, representing solution quality and enabling correctness checks across different models.
- **Number of branch-and-bound nodes**, which quantifies the size of the explored search tree. This metric is a direct indicator of combinatorial difficulty and solver effort.

9.2 CP Performance Metrics

Constraint Programming relies on propagation and search rather than LP relaxations. As a result, performance is best understood through metrics that describe search dynamics and propagation strength.

- **Solution status**, indicating whether optimality, feasibility, or infeasibility was reached.

- **Number of conflicts**, capturing how often the solver encounters contradictions during search; high values typically indicate weak propagation or difficult constraint interactions.
- **Number of branches**, representing the size of the explored decision tree.
- **Number of Boolean variables**, which reflects the combinatorial structure of the model.
- **Best objective bound**, useful for assessing solution quality when optimality is not proven.
- **Number of variables** and **constraints**, allowing structural comparison with the MIP formulation.

9.3 Hybrid Model Performance Metrics

The hybrid CP–LP approach combines two fundamentally different solving paradigms, requiring a more detailed and decomposed performance analysis.

The hybrid metrics are organized into three categories:

Global metrics
- **Total wall-clock time**, measuring end-to-end performance.
- **Number of LBBD iterations**, indicating how many master–subproblem interactions were required.
- **Convergence flag**, identifying whether the algorithm reached a fixed point where master and subproblem costs agree.

CP (Master Problem) metrics
- **Total CP time**, capturing the cumulative time spent solving the master problem.
- **Number of branches** and **conflicts**, describing the combinatorial search effort.
- **Number of Boolean variables**, variables, and constraints, characterizing master problem size and complexity.

LP (Subproblem) metrics
- **Number of LP calls**, corresponding to the number of Benders iterations.
- **Total LP time**, measuring the continuous optimization effort.

In addition, **peak memory usage** is tracked throughout the entire hybrid execution to capture the overhead introduced by maintaining both CP and LP components simultaneously.

9.4 Role of Performance Metrics in Model Comparison

The chosen metrics enable both intra-model analysis and cross-model comparison. Execution time and memory usage provide a direct measure of practical efficiency, while search-related metrics (branches, conflicts, nodes, iterations) explain why one approach scales better than another.

In particular:
- MIP metrics highlight the rapid growth of combinatorial complexity.
- CP metrics reveal the impact of search strategies and propagation strength.
- Hybrid metrics demonstrate how decomposition redistributes complexity and improves scalability.

Together, these performance indicators provide a comprehensive framework for evaluating solution quality, computational effort, and scalability across all three modeling approaches.

10 Performance Evaluation and Comparative Analysis

10.1 Temporal Performance and Stability

The overall analysis of computational performance identifies the number of runways as the critical parameter defining problem complexity. As illustrated in Fig. 2, for small-scale instances (one or two runways), all methods exhibit reduced execution times. However, as the number of runways increases, a clear divergence in solver behavior is observed. The CP approach demonstrates the greatest stability, with a gradual and predictable growth in execution time, remaining consistently superior to the other approaches up to six runways. In contrast, MIP exhibits irregular behavior, characterized by abrupt increases in average execution time and high variability, particularly noticeable from four runways onwards. The Hybrid approach, while competitive in small instances, suffers severe performance degradation from six runways onwards, becoming the least efficient option for complex scenarios.

This trend is corroborated by the heatmap analysis in Fig. 3, where the logarithmic scale highlights that multiple-runway instances impose a significantly higher computational cost than single-runway ones. The lighter colors associated with MIP and Hybrid in larger datasets confirm their difficulty in handling the problem's combinatorial structure, while CP maintains consistent performance (evidenced by darker colors). Furthermore, the statistical analysis presented in Figure 8 reinforces CP's superiority regarding predictability. The boxplots demonstrate that CP maintains low dispersion and low medians, whereas MIP and Hybrid present not only higher medians but also a significant presence of outliers, indicating a lack of robustness in the face of instance variability.

10.2 Structural Complexity of the Models

The disparity in execution times can be largely attributed to the internal formulation of the models generated by each solver, as detailed in Fig. 4. In smaller datasets, the Hybrid model systematically generates the highest number of variables and constraints, foreshadowing its scaling difficulties. As the dataset dimension increases, an almost explosive growth in the number of constraints is observed for the MIP model, reaching values that far exceed those of other approaches. Conversely, CP is distinguished by its structural compactness, maintaining the lowest number of variables and constraints across all analyzed scenarios. This efficiency in model formulation is a determining factor for the observed scalability, avoiding the computational overload that penalizes the performance of MIP and Hybrid.

10.3 Trade-off between Execution Time and Memory Consumption

However, the temporal efficiency of CP is not without cost, presenting a clear trade-off between speed and memory consumption. Fig. 5 and Fig. 7 illustrate this compromise: while CP consistently offers the lowest execution times, its memory consumption grows progressively, making it the most resource-demanding approach for large-scale instances with multiple runways. On the other hand, MIP and Hybrid maintain relatively low and stable memory consumption levels, even in large instances. The global efficiency analysis (Fig.7) positions CP in the region of best temporal performance, albeit with greater dispersion along the memory axis, while MIP scatters along the time axis without offering resource savings that justify its slowness. The Hybrid approach frequently occupies an unfavorable region, combining intermediate or high times with considerable memory consumption.

10.4 Scalability and Search Effort

The scalability analysis, depicted in Fig. 10, demonstrates that CP exhibits the most favorable relationship between problem complexity (measured by the number of variables and constraints) and execution time. The time growth in CP is controlled, whereas MIP shows extreme sensitivity to dimensional increases, with times scaling abruptly. This behavior is intrinsically linked to the search effort detailed in Fig. 9. MIP is forced to explore an excessively high number of nodes to prove optimality in large instances, reflecting its difficulty in pruning the solution space. In contrast, CP generates a significantly lower number of branches, indicating a more efficient and targeted exploration of the search space. The hybrid approach fails to buffer this impact, inheriting the combinatorial complexity of MIP without effectively reducing the global search space.

10.5 Analysis of Internal Mechanisms and Overhead

A deeper investigation into the time components reveals the structural causes behind the poor performance of the hybrid approach. Fig. 6 decomposes the total time of the Hybrid solver, evidencing that in large-scale instances, the coordination overhead between the CP and MIP paradigms becomes non-negligible. The integration cost outweighs the potential benefits of cooperation, resulting in performance inferior to the sum of its parts. Regarding CP's internal mechanisms, Fig. 11 and 12 show a healthy balance between decision tree exploration (branches) and conflict learning (conflicts). Although the number of auxiliary boolean variables grows with problem dimension, this increase is proportional and controlled, allowing CP's propagation and learning mechanisms to maintain effectiveness even when the search relies more on branching than on logical deduction.

10.6 Sensitivity to Search Strategies

The efficiency of the resolution process is often contingent upon the specific mechanisms used to navigate the search space. Fig. 13 evaluates the impact of different search strategies (Automatic, Fixed, LP, Portfolio) on final performance. The results indicate that the choice of strategy has a marginal impact when compared to the choice of the resolution paradigm. In the case of CP, strategies such as Automatic and Portfolio offer slight advantages in larger instances but do not alter the order of magnitude of execution time. For the Hybrid approach, the influence of the strategy is practically null, suggesting that this approach's limitations are fundamentally structural and cannot be corrected through search parameterization. Thus, it is concluded that the robustness of the CP model is intrinsic to its formulation and ability to handle logical constraints, superseding the specific tuning of search algorithms.

11 Conclusion

This study has presented a comprehensive comparative analysis of three distinct computational paradigms—Mixed-Integer Programming (MIP), Constraint Programming (CP), and Hybrid Logic-Based Benders Decomposition (LBBD), applied to the static Aircraft Landing Problem (ALP). By systematically formulating and testing these models across a spectrum of benchmark instances, ranging from simple single-runway scenarios to complex multi-runway environments, this research has delineated the operational boundaries and scalability of each approach.

The experimental evaluation confirms that while all three formulations successfully converge to correct optimal solutions (validating the mathematical integrity of the proposed models), their computational efficiency diverges significantly as problem dimensions increase. The investigation identified the number of aircraft and, crucially, the number of runways as the dominant parameters governing problem complexity. While the MIP formulation proved effective for small-scale instances, it exhibited severe limitations in scalability. The exponential growth in constraints and the necessity to explore a massive branch-and-bound tree rendered MIP computationally prohibitive for large, multi-runway datasets. Similarly, the Hybrid LBBD approach, despite its theoretical elegance in decomposing combinatorial and continuous decisions, struggled with the overhead of master-subproblem coordination, often yielding performance metrics inferior to the monolithic models.

In sharp contrast, Constraint Programming emerged as the most robust and efficient paradigm. The CP model demonstrated superior scalability, maintaining stable execution times and effective search capabilities even in the most demanding instances. Although this temporal efficiency incurs a higher memory cost, the trade-off is justifiable given the critical requirement for rapid decision-making in air traffic control. Furthermore, the analysis of search strategies revealed that while heuristic tuning (such as Portfolio or LP-guided search) can offer marginal improvements, the intrinsic propagation strength of the CP solver is the primary determinant of success. The fundamental ability of CP to handle logical disjunctions and non-linear constraints naturally outweighs the benefits of specific branching heuristics.

Ultimately, this research establishes that for the static Aircraft Landing Problem, the choice of the modeling paradigm is far more consequential than algorithmic fine-tuning. The dominance of Constraint Programming suggests that future developments in real-time Air Traffic Management systems should prioritize constraint-based architectures. By proving that CP can deliver optimal schedules with high predictability and low latency, this work provides a solid foundation for the development of next-generation decision support tools capable of handling the increasing density and complexity of global airspace.

References

1. Beasley, J.E., Krishnamoorthy, M., Sharaiha, Y.M., Abramson, D.: Scheduling aircraft landings – the static case. December 1995, revised February 1998, last accessed 2026/01/12
2. Ciré, A. A., Çoban, E., & Hooker, J. N. (2004). Logic-based Benders decomposition for planning and scheduling: A computational analysis. The Knowledge Engineering Review, 1–24, last accessed 2026/01/16, available on https://johnhooker.tepper.cmu.edu/bendersKER.pdf

Appendix

Table 1. Estimated growth of MIP model complexity and runtime.

Dataset	Aircraft	Dec. Vars	Constraints	Est. Runtime
9	100	10,200	12,019	Days
10	150	22,800	25,569	Days / Weeks
11	200	40,400	44,184	Weeks / Months
12	250	63,000	67,977	Months / Years
13	500	251,000	261,552	**Impractical**

Table 2. Runway configurations tested for large datasets.

Dataset	Runways Tested
9	3 – 4
10	4 – 5
11	5
12	6
13	7

Additional Visualizations for the Result Analysis

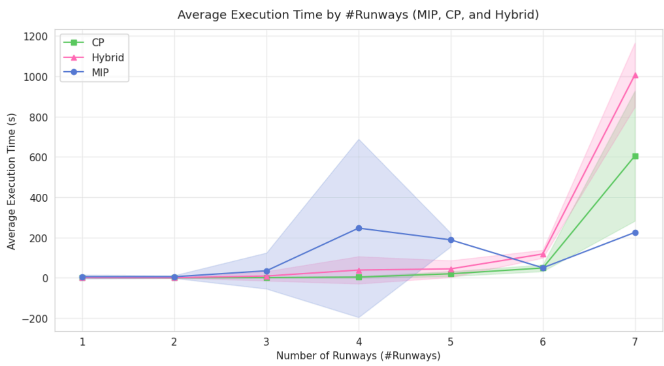

Fig. 2. Average Execution Time by Runways per Solver

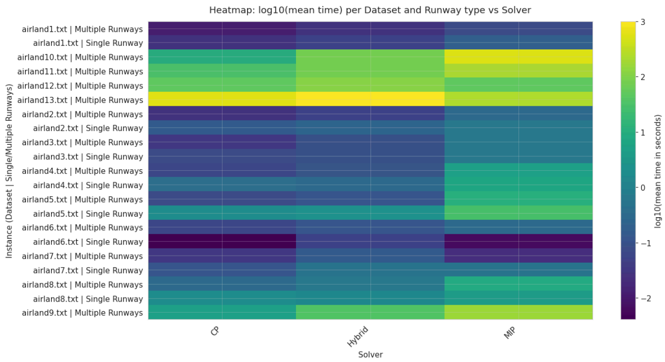

Fig. 3. Heatmap log10(mean time) per Dataset and Runway type vs Solver

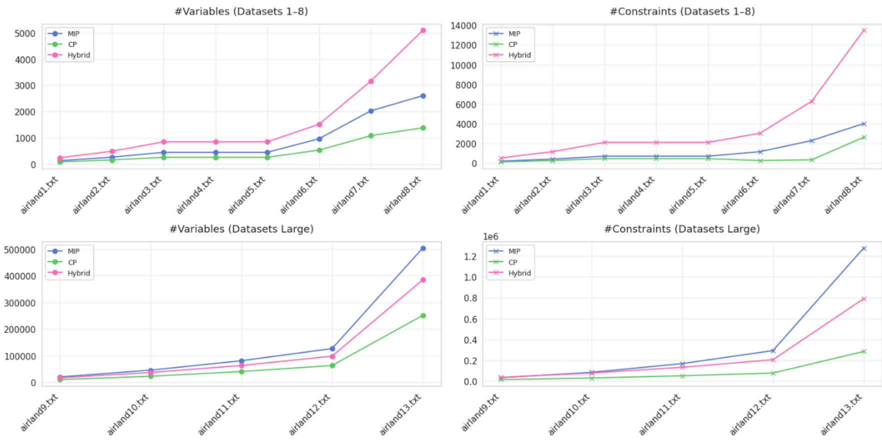

Fig. 4. Number of Variables and Constraints per Solver

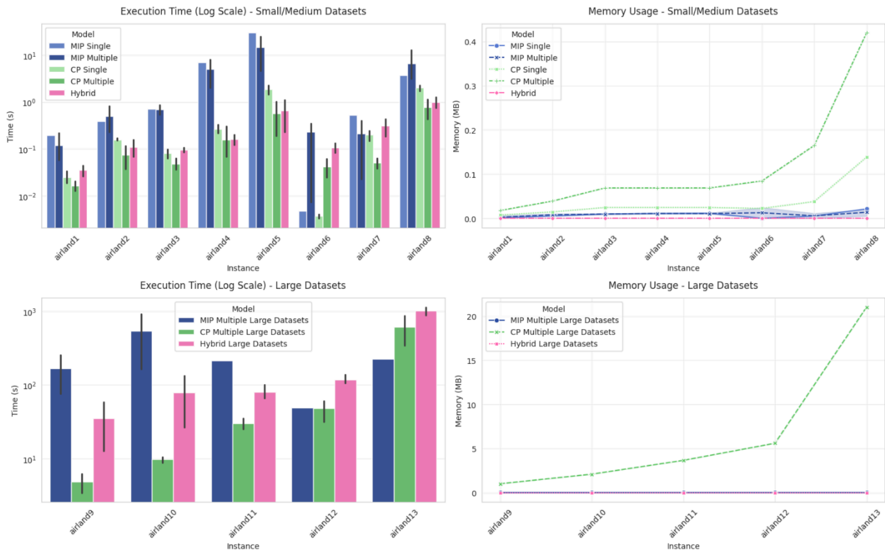

Fig. 5. Overall Comparison of Time and Memory

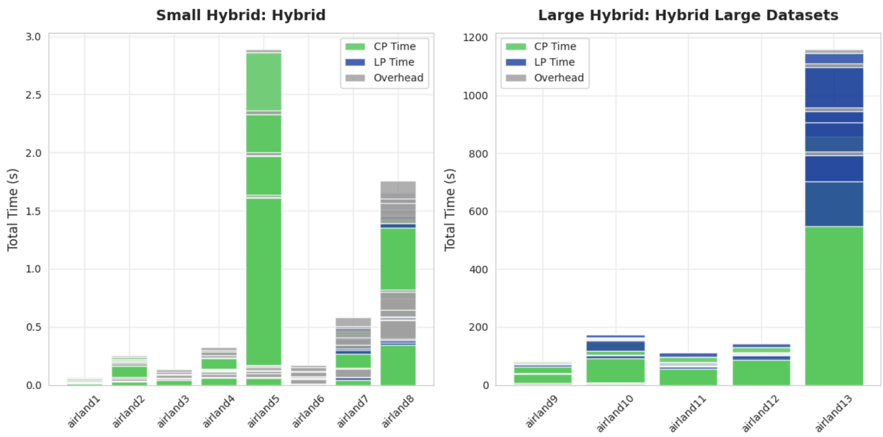

Fig. 6. Execution Time Hybrid

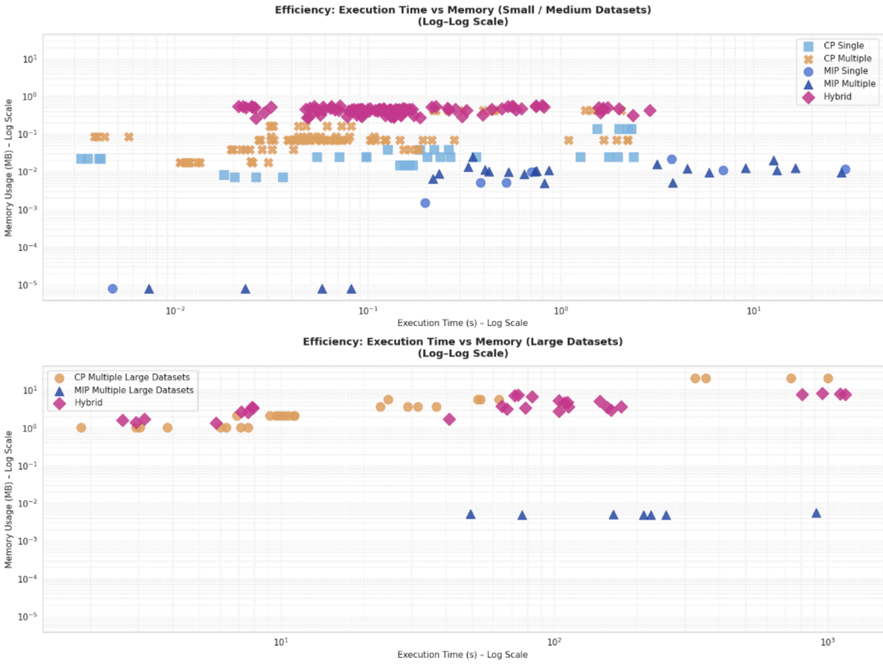

Fig. 7. Efficiency between models

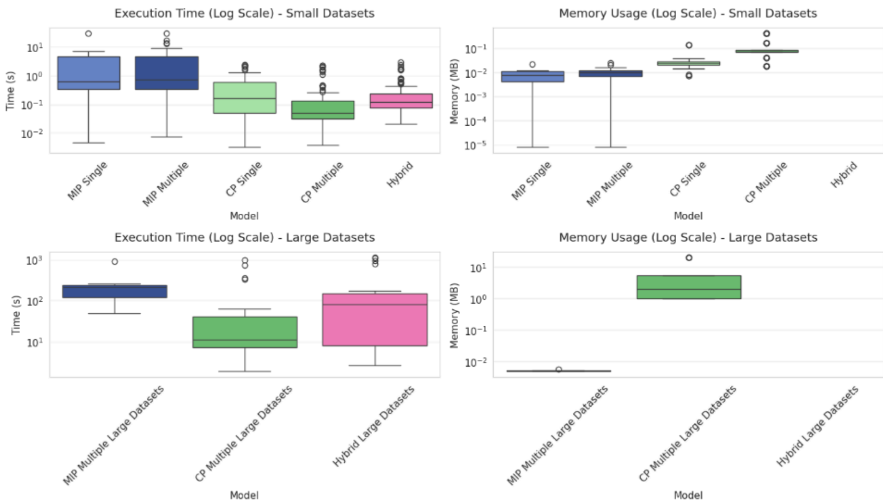

Fig. 8. Comparison in Boxplots of Execution Time and Memory Usage

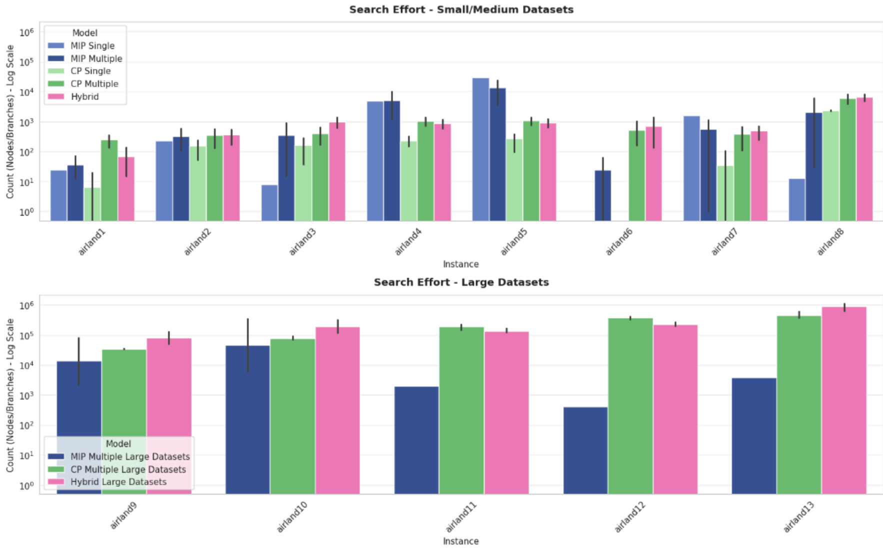

Fig. 9. Search Effort: MIP Nodes vs CP Branches

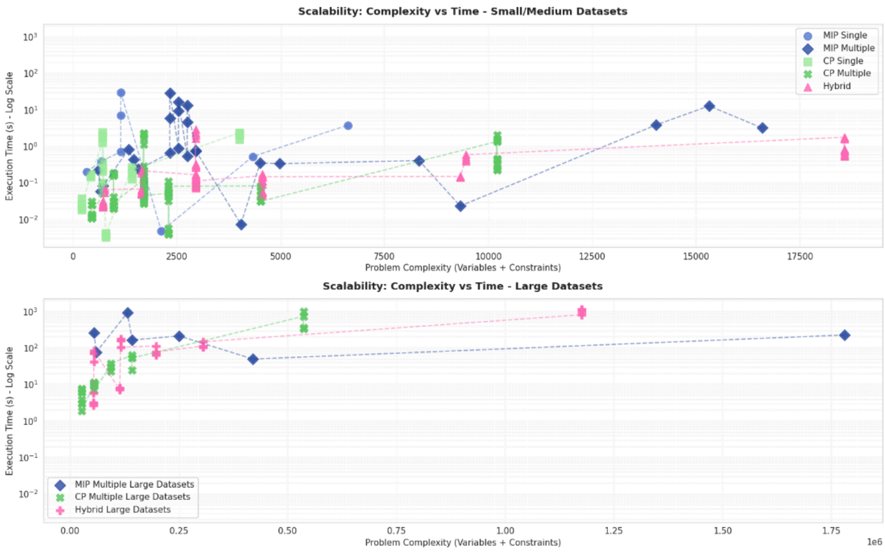

Fig. 10. Scalability: Complexity vs Time

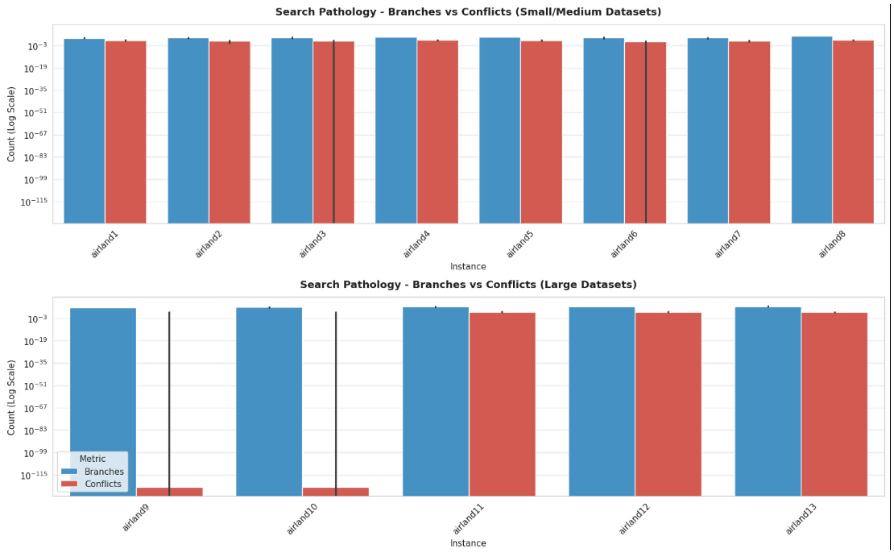

Fig. 11. CP: Search Pathology - Decision Tree (Branches) vs Learning (Conflicts)

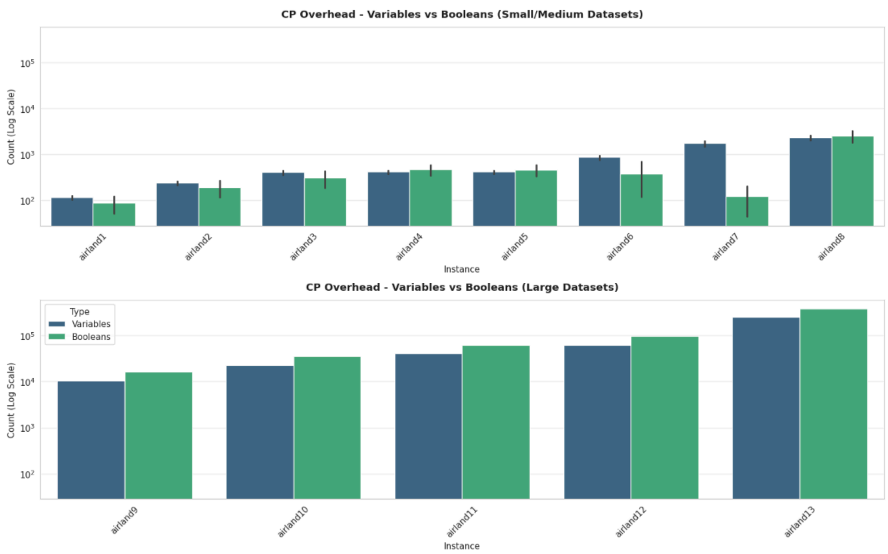

Fig. 12. CP Overhead - Variables vs Booleans

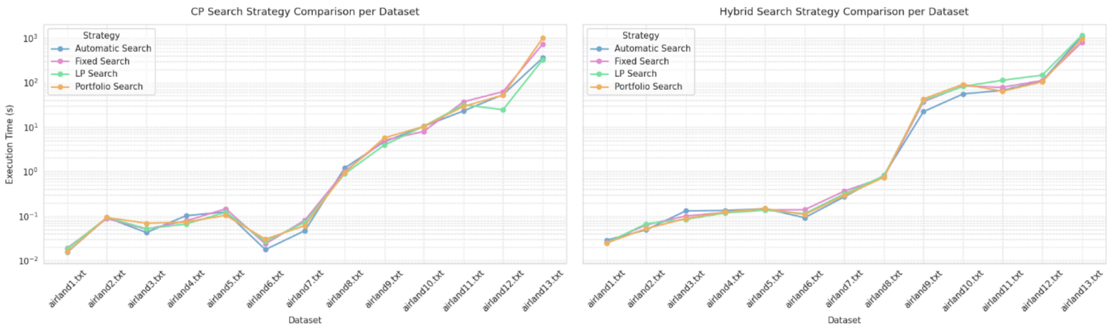

Fig. 13. CP and Hybrid: Strategy Comparison per Dataset

Results for each Dataset

Interactive Metrics Table

Filter by Dataset: × airland1.txt

Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland1.txt	10	1	MIP Single	–	0.19742178916931152	0.001492	700	120	205	25	–	–	–	–	–
airland1.txt	10	1	MIP Multiple	–	0.21750473976135254	0.006609	700	220	395	21	–	–	–	–	–
airland1.txt	10	2	MIP Multiple	–	0.057691097259521484	0.000008	90	230	440	73	–	–	–	–	–
airland1.txt	10	3	MIP Multiple	–	0.08179664611816406	0.000008	0	240	485	13	–	–	–	–	–
airland1.txt	10	1	CP Single	Automatic Search	0.0179205	0.008193	700	75	130	0	OPTIMAL	0	0	–	–
airland1.txt	10	1	CP Single	Fixed Search	0.0359976	0.007297	700	75	130	0	OPTIMAL	0	0	–	–
airland1.txt	10	1	CP Single	Portfolio Search	0.0202635	0.007297	700	75	130	0	OPTIMAL	0	0	–	–
airland1.txt	10	1	CP Single	LP Search	0.0261805	0.007297	700	75	130	26	OPTIMAL	0	45	–	–
airland1.txt	10	1	CP Multiple	Automatic Search	0.0246698	0.018498	700	130	320	0	OPTIMAL	0	45	–	–
airland1.txt	10	1	CP Multiple	Fixed Search	0.025167	0.017602	700	130	320	98	OPTIMAL	0	53	–	–
airland1.txt	10	1	CP Multiple	Portfolio Search	0.0303954	0.017602	700	130	320	0	OPTIMAL	0	45	–	–
airland1.txt	10	1	CP Multiple	LP Search	0.0250298	0.017602	700	130	320	84	OPTIMAL	0	45	–	–
airland1.txt	10	2	CP Multiple	Automatic Search	0.0119134	0.017602	90	130	320	196	OPTIMAL	0	100	–	–
airland1.txt	10	2	CP Multiple	Fixed Search	0.0134362	0.017602	90	130	320	337	OPTIMAL	13	120	–	–
airland1.txt	10	2	CP Multiple	Portfolio Search	0.0105931	0.017602	90	130	320	0	OPTIMAL	0	0	–	–
airland1.txt	10	2	CP Multiple	LP Search	0.011713	0.017602	90	130	320	427	OPTIMAL	19	131	–	–
airland1.txt	10	3	CP Multiple	Automatic Search	0.0129762	0.017602	0	130	320	360	OPTIMAL	0	180	–	–
airland1.txt	10	3	CP Multiple	Fixed Search	0.0106335	0.017602	0	130	320	479	OPTIMAL	9	210	–	–
airland1.txt	10	3	CP Multiple	Portfolio Search	0.0115664	0.017602	0	130	320	479	OPTIMAL	9	210	–	–
airland1.txt	10	3	CP Multiple	LP Search	0.0114362	0.017602	0	130	320	479	OPTIMAL	9	210	–	–
airland1.txt	10	1	Hybrid	Automatic Search	0.0548518	0.4175224	700	232	529	92	true	0	47	2	2
airland1.txt	10	1	Hybrid	Fixed Search	0.0312684	0.5188713	700	221	506	0	true	0	0	1	1
airland1.txt	10	1	Hybrid	Portfolio Search	0.0646918	0.5322723	700	232	529	92	true	0	47	2	2
airland1.txt	10	1	Hybrid	LP Search	0.0714331	0.5574532	700	232	529	60	true	0	47	2	2
airland1.txt	10	2	Hybrid	Automatic Search	0.0288806	0.3664055	90	221	506	0	true	0	145	1	1
airland1.txt	10	2	Hybrid	Fixed Search	0.0232315	0.4975967	90	221	506	363	true	0	163	1	1
airland1.txt	10	2	Hybrid	Portfolio Search	0.0224059	0.5322723	90	221	506	0	true	0	0	1	1
airland1.txt	10	2	Hybrid	LP Search	0.0212293	0.5391541	90	221	506	0	true	0	0	1	1
airland1.txt	10	3	Hybrid	Automatic Search	0.0261891	0.2609825	0	221	506	0	true	0	0	1	1
airland1.txt	10	3	Hybrid	Fixed Search	0.025763	0.4975967	0	221	506	234	true	0	180	1	1
airland1.txt	10	3	Hybrid	Portfolio Search	0.0248353	0.5322723	0	221	506	0	true	0	0	1	1
airland1.txt	10	3	Hybrid	LP Search	0.0250778	0.5227089	0	221	506	0	true	0	0	1	1

Fig. 14. Results for Dataset 1

Interactive Metrics Table

Filter by Dataset: × airland2.txt

Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland2.txt	15	1	MIP Single	–	0.3837614059448242	0.005136	1480	255	420	234	–	–	–	–	–
airland2.txt	15	1	MIP Multiple	–	0.8208167552947998	0.004982	1480	480	855	256	–	–	–	–	–
airland2.txt	15	2	MIP Multiple	–	0.4245872497558594	0.010298	210	495	960	588	–	–	–	–	–
airland2.txt	15	3	MIP Multiple	–	0.2332170009613037	0.008766	0	510	1065	109	–	–	–	–	–
airland2.txt	15	1	CP Single	Automatic Search	0.1635006	0.014567	1480	150	270	271	OPTIMAL	0	159	–	–
airland2.txt	15	1	CP Single	Fixed Search	0.1703024	0.014567	1480	150	270	132	OPTIMAL	0	105	–	–
airland2.txt	15	1	CP Single	Portfolio Search	0.1457854	0.014567	1480	150	270	210	OPTIMAL	0	105	–	–
airland2.txt	15	1	CP Single	LP Search	0.1555566	0.014567	1480	150	270	0	OPTIMAL	0	0	–	–
airland2.txt	15	1	CP Multiple	Automatic Search	0.1826077	0.039057	1480	270	705	78	OPTIMAL	0	105	–	–
airland2.txt	15	1	CP Multiple	Fixed Search	0.1530293	0.039057	1480	270	705	239	OPTIMAL	0	127	–	–
airland2.txt	15	1	CP Multiple	Portfolio Search	0.1803761	0.039057	1480	270	705	98	OPTIMAL	0	105	–	–
airland2.txt	15	1	CP Multiple	LP Search	0.1663588	0.039057	1480	270	705	158	OPTIMAL	0	105	–	–
airland2.txt	15	2	CP Multiple	Automatic Search	0.019483	0.039057	210	270	705	0	OPTIMAL	0	0	–	–
airland2.txt	15	2	CP Multiple	Fixed Search	0.0199007	0.039057	210	270	705	544	OPTIMAL	1	255	–	–
airland2.txt	15	2	CP Multiple	Portfolio Search	0.0410226	0.039057	210	270	705	0	OPTIMAL	0	225	–	–
airland2.txt	15	2	CP Multiple	LP Search	0.0242627	0.039057	210	270	705	0	OPTIMAL	0	0	–	–
airland2.txt	15	3	CP Multiple	Automatic Search	0.0219841	0.039057	0	270	705	930	OPTIMAL	4	450	–	–
airland2.txt	15	3	CP Multiple	Fixed Search	0.0230328	0.039057	0	270	705	74	OPTIMAL	0	420	–	–
airland2.txt	15	3	CP Multiple	Portfolio Search	0.0282426	0.039057	0	270	705	1044	OPTIMAL	10	450	–	–
airland2.txt	15	3	CP Multiple	LP Search	0.0316801	0.039057	0	270	705	1006	OPTIMAL	10	450	–	–
airland2.txt	15	1	Hybrid	Automatic Search	0.1859705	0.2692146	1480	481	1171	336	true	0	126	1	1
airland2.txt	15	1	Hybrid	Fixed Search	0.2575493	0.4844398	1480	481	1171	140	true	0	105	1	1
airland2.txt	15	1	Hybrid	Portfolio Search	0.2240605	0.5322723	1480	481	1171	246	true	0	105	1	1
airland2.txt	15	1	Hybrid	LP Search	0.2139847	0.5228577	1480	481	1171	0	true	0	0	1	1
airland2.txt	15	2	Hybrid	Automatic Search	0.0485914	0.2759285	210	481	1171	761	true	0	330	1	1
airland2.txt	15	2	Hybrid	Fixed Search	0.0495961	0.4844398	210	481	1171	808	true	0	358	1	1
airland2.txt	15	2	Hybrid	Portfolio Search	0.0523524	0.5321159	210	481	1171	720	true	0	330	1	1
airland2.txt	15	2	Hybrid	LP Search	0.0585036	0.5228577	210	481	1171	775	true	0	330	1	1
airland2.txt	15	3	Hybrid	Automatic Search	0.0492272	0.277298	0	481	1171	0	true	0	420	1	1
airland2.txt	15	3	Hybrid	Fixed Search	0.0635743	0.4844398	0	481	1171	556	true	0	420	1	1
airland2.txt	15	3	Hybrid	Portfolio Search	0.0524457	0.4400024	0	481	1171	0	true	0	0	1	1
airland2.txt	15	3	Hybrid	LP Search	0.0673048	0.4360695	0	481	1171	0	true	0	0	1	1

Fig. 15. Results for Dataset 2

Interactive Metrics Table

Filter by Dataset: × airland3.txt

Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland3.txt	20	1	MIP Single	–	0.7027649879455566	0.009821	820	440	710	8	–	–	–	–	–
airland3.txt	20	1	MIP Multiple	–	0.6474666595458984	0.008446	820	840	1490	15	–	–	–	–	–
airland3.txt	20	2	MIP Multiple	–	0.8670163154602051	0.010969	60	860	1680	912	–	–	–	–	–
airland3.txt	20	3	MIP Multiple	–	0.5358476638793945	0.009655	0	880	1870	107	–	–	–	–	–
airland3.txt	20	1	CP Single	Automatic Search	0.054101	0.024524	820	250	460	0	OPTIMAL	0	0	–	–
airland3.txt	20	1	CP Single	Fixed Search	0.0979231	0.024524	820	250	460	320	OPTIMAL	0	190	–	–
airland3.txt	20	1	CP Single	Portfolio Search	0.097814	0.024524	820	250	460	256	OPTIMAL	0	190	–	–
airland3.txt	20	1	CP Single	LP Search	0.0708132	0.024524	820	250	460	74	OPTIMAL	0	190	–	–
airland3.txt	20	1	CP Multiple	Automatic Search	0.0438881	0.068676	820	460	1240	378	OPTIMAL	0	190	–	–
airland3.txt	20	1	CP Multiple	Fixed Search	0.056961	0.068676	820	460	1240	116	OPTIMAL	0	190	–	–
airland3.txt	20	1	CP Multiple	Portfolio Search	0.12205	0.068676	820	460	1240	254	OPTIMAL	0	190	–	–
airland3.txt	20	1	CP Multiple	LP Search	0.0624249	0.068676	820	460	1240	394	OPTIMAL	0	204	–	–
airland3.txt	20	2	CP Multiple	Automatic Search	0.0312547	0.068676	60	460	1240	0	OPTIMAL	0	400	–	–
airland3.txt	20	2	CP Multiple	Fixed Search	0.026969	0.068676	60	460	1240	404	OPTIMAL	0	400	–	–
airland3.txt	20	2	CP Multiple	Portfolio Search	0.0275571	0.068676	60	460	1240	0	OPTIMAL	0	0	–	–
airland3.txt	20	2	CP Multiple	LP Search	0.0324837	0.068676	60	460	1240	1453	OPTIMAL	17	462	–	–
airland3.txt	20	3	CP Multiple	Automatic Search	0.0417185	0.068676	0	460	1240	0	OPTIMAL	0	0	–	–
airland3.txt	20	3	CP Multiple	Fixed Search	0.0451289	0.068676	0	460	1240	948	OPTIMAL	0	760	–	–
airland3.txt	20	3	CP Multiple	Portfolio Search	0.0402202	0.068676	0	460	1240	0	OPTIMAL	0	760	–	–
airland3.txt	20	3	CP Multiple	LP Search	0.0415301	0.068676	0	460	1240	818	OPTIMAL	0	760	–	–
airland3.txt	20	1	Hybrid	Automatic Search	0.1358786	0.277298	820	841	2111	0	true	0	0	1	1
airland3.txt	20	1	Hybrid	Fixed Search	0.1008976	0.4103203	820	841	2111	380	true	0	190	1	1
airland3.txt	20	1	Hybrid	Portfolio Search	0.0910721	0.4400024	820	841	2111	0	true	0	0	1	1
airland3.txt	20	1	Hybrid	LP Search	0.0895109	0.4398842	820	841	2111	0	true	0	0	1	1
airland3.txt	20	2	Hybrid	Automatic Search	0.0779552	0.2875633	60	841	2111	1200	true	0	590	1	1
airland3.txt	20	2	Hybrid	Fixed Search	0.0792532	0.4165764	60	841	2111	1426	true	0	626	1	1
airland3.txt	20	2	Hybrid	Portfolio Search	0.0704985	0.4284134	60	841	2111	1353	true	0	590	1	1
airland3.txt	20	2	Hybrid	LP Search	0.0819747	0.4398842	60	841	2111	814	true	0	590	1	1
airland3.txt	20	3	Hybrid	Automatic Search	0.1312561	0.2875633	0	841	2111	1747	true	0	761	1	1
airland3.txt	20	3	Hybrid	Fixed Search	0.1121309	0.4165764	0	841	2111	1783	true	0	796	1	1
airland3.txt	20	3	Hybrid	Portfolio Search	0.0879912	0.4383011	0	841	2111	1649	true	0	760	1	1
airland3.txt	20	3	Hybrid	LP Search	0.0846534	0.4539223	0	841	2111	1616	true	0	760	1	1

Fig. 16. Results for Dataset 3

Interactive Metrics Table

Filter by Dataset: airland4.txt

Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland4.txt	20	1	MIP Single	–	6.932842969894409	0.010872	2520	440	710	4856	–	–	–	–	–
airland4.txt	20	1	MIP Multiple	–	5.8729212284088135	0.009341	2520	840	1490	4456	–	–	–	–	–
airland4.txt	20	2	MIP Multiple	–	9.107760190963745	0.012292	640	860	1680	12868	–	–	–	–	–
airland4.txt	20	3	MIP Multiple	–	4.521025657653809	0.011878	130	880	1870	2451	–	–	–	–	–
airland4.txt	20	4	MIP Multiple	–	0.7330935001373291	0.010103	0	900	2060	175	–	–	–	–	–
airland4.txt	20	1	CP Single	Automatic Search	0.3633198	0.024519	2520	250	460	202	OPTIMAL	0	190	–	–
airland4.txt	20	1	CP Single	Fixed Search	0.2670201	0.024519	2520	250	460	368	OPTIMAL	0	190	–	–
airland4.txt	20	1	CP Single	Portfolio Search	0.2024201	0.024519	2520	250	460	198	OPTIMAL	0	190	–	–
airland4.txt	20	1	CP Single	LP Search	0.2367733	0.024519	2520	250	460	134	OPTIMAL	0	190	–	–
airland4.txt	20	1	CP Multiple	Automatic Search	0.1919051	0.068667	2520	460	1240	380	OPTIMAL	0	190	–	–
airland4.txt	20	1	CP Multiple	Fixed Search	0.2788845	0.068667	2520	460	1240	174	OPTIMAL	0	190	–	–
airland4.txt	20	1	CP Multiple	Portfolio Search	1.0982789	0.068667	2520	460	1240	0	OPTIMAL	0	0	–	–
airland4.txt	20	1	CP Multiple	LP Search	0.2145962	0.068667	2520	460	1240	0	OPTIMAL	0	190	–	–
airland4.txt	20	2	CP Multiple	Automatic Search	0.1028251	0.068667	640	460	1240	76	OPTIMAL	0	400	–	–
airland4.txt	20	2	CP Multiple	Fixed Search	0.0445679	0.068667	640	460	1240	1352	OPTIMAL	22	467	–	–
airland4.txt	20	2	CP Multiple	Portfolio Search	0.0738884	0.068667	640	460	1240	1184	OPTIMAL	15	440	–	–
airland4.txt	20	2	CP Multiple	LP Search	0.0479442	0.068667	640	460	1240	1352	OPTIMAL	22	467	–	–
airland4.txt	20	3	CP Multiple	Automatic Search	0.0682238	0.068667	130	460	1240	1615	OPTIMAL	5	800	–	–
airland4.txt	20	3	CP Multiple	Fixed Search	0.0780387	0.068667	130	460	1240	1396	OPTIMAL	0	760	–	–
airland4.txt	20	3	CP Multiple	Portfolio Search	0.0661323	0.068667	130	460	1240	1783	OPTIMAL	26	800	–	–
airland4.txt	20	3	CP Multiple	LP Search	0.0666324	0.068667	130	460	1240	1310	OPTIMAL	0	760	–	–
airland4.txt	20	4	CP Multiple	Automatic Search	0.0530837	0.068667	0	460	1240	1870	OPTIMAL	18	800	–	–
airland4.txt	20	4	CP Multiple	Fixed Search	0.0463708	0.068667	0	460	1240	1236	OPTIMAL	0	760	–	–
airland4.txt	20	4	CP Multiple	Portfolio Search	0.0387351	0.068667	0	460	1240	1870	OPTIMAL	18	800	–	–
airland4.txt	20	4	CP Multiple	LP Search	0.0382423	0.068667	0	460	1240	1228	OPTIMAL	0	760	–	–
airland4.txt	20	1	Hybrid	Automatic Search	0.307775	0.2931976	2520	841	2111	597	true	0	217	1	1
airland4.txt	20	1	Hybrid	Fixed Search	0.3240786	0.4223862	2520	841	2111	0	true	0	190	1	1
airland4.txt	20	1	Hybrid	Portfolio Search	0.261615	0.4419556	2520	841	2111	597	true	0	217	1	1
airland4.txt	20	1	Hybrid	LP Search	0.2835104	0.4539223	2520	841	2111	0	true	0	0	1	1
airland4.txt	20	2	Hybrid	Automatic Search	0.1357086	0.2947006	640	841	2111	0	true	0	0	1	1
airland4.txt	20	2	Hybrid	Fixed Search	0.1114726	0.4223862	640	841	2111	1352	true	0	590	1	1
airland4.txt	20	2	Hybrid	Portfolio Search	0.1283088	0.4419556	640	841	2111	1427	true	0	627	1	1
airland4.txt	20	2	Hybrid	LP Search	0.1209512	0.4581871	640	841	2111	416	true	0	590	1	1
airland4.txt	20	3	Hybrid	Automatic Search	0.1320639	0.3091965	130	841	2111	880	true	0	760	1	1
airland4.txt	20	3	Hybrid	Fixed Search	0.1305768	0.4283752	130	841	2111	922	true	0	760	1	1
airland4.txt	20	3	Hybrid	Portfolio Search	0.1207175	0.4489937	130	841	2111	760	true	0	760	1	1
airland4.txt	20	3	Hybrid	LP Search	0.1151979	0.4581871	130	841	2111	1182	true	0	760	1	1
airland4.txt	20	4	Hybrid	Automatic Search	0.0918379	0.3119202	0	841	2111	1803	true	2	797	1	1
airland4.txt	20	4	Hybrid	Fixed Search	0.0939322	0.4283752	0	841	2111	1803	true	2	797	1	1
airland4.txt	20	4	Hybrid	Portfolio Search	0.0997763	0.4489937	0	841	2111	718	true	0	760	1	1
airland4.txt	20	4	Hybrid	LP Search	0.0932958	0.4619942	0	841	2111	1348	true	0	760	1	1

Fig. 17. Results for Dataset 4

Interactive Metrics Table

Filter by Dataset: airland5.txt

Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland5.txt	20	1	MIP Single	–	29.85306406021118	0.011539	3100	440	710	29506	–	–	–	–	–
airland5.txt	20	1	MIP Multiple	–	28.545490741729736	0.009341	3100	840	1490	28876	–	–	–	–	–
airland5.txt	20	2	MIP Multiple	–	16.499996423721313	0.012516	650	860	1680	17387	–	–	–	–	–
airland5.txt	20	3	MIP Multiple	–	13.233667135238647	0.010763	170	880	1870	6979	–	–	–	–	–
airland5.txt	20	4	MIP Multiple	–	0.7466866970062256	0.010436	0	900	2060	159	–	–	–	–	–
airland5.txt	20	1	CP Single	Automatic Search	1.7732932	0.024519	3100	250	460	390	OPTIMAL	0	200	–	–
airland5.txt	20	1	CP Single	Fixed Search	1.9697098	0.024519	3100	250	460	300	OPTIMAL	0	190	–	–
airland5.txt	20	1	CP Single	Portfolio Search	2.3820776	0.024519	3100	250	460	386	OPTIMAL	0	197	–	–
airland5.txt	20	1	CP Single	LP Search	1.2597007	0.024519	3100	250	460	0	OPTIMAL	0	190	–	–
airland5.txt	20	1	CP Multiple	Automatic Search	2.2184674	0.068667	3100	460	1240	224	OPTIMAL	0	190	–	–
airland5.txt	20	1	CP Multiple	Fixed Search	2.2322762	0.068667	3100	460	1240	479	OPTIMAL	0	240	–	–
airland5.txt	20	1	CP Multiple	Portfolio Search	1.6653118	0.068667	3100	460	1240	92	OPTIMAL	0	190	–	–
airland5.txt	20	1	CP Multiple	LP Search	1.9429284	0.068667	3100	460	1240	402	OPTIMAL	0	212	–	–
airland5.txt	20	2	CP Multiple	Automatic Search	0.1240234	0.068667	650	460	1240	1250	OPTIMAL	12	440	–	–
airland5.txt	20	2	CP Multiple	Fixed Search	0.0575946	0.068667	650	460	1240	1020	OPTIMAL	1	440	–	–
airland5.txt	20	2	CP Multiple	Portfolio Search	0.0604115	0.068667	650	460	1240	0	OPTIMAL	0	0	–	–
airland5.txt	20	2	CP Multiple	LP Search	0.0718113	0.068667	650	460	1240	1111	OPTIMAL	7	440	–	–
airland5.txt	20	3	CP Multiple	Automatic Search	0.109914	0.068667	170	460	1240	1720	OPTIMAL	10	800	–	–
airland5.txt	20	3	CP Multiple	Fixed Search	0.1464469	0.068667	170	460	1240	1514	OPTIMAL	0	760	–	–
airland5.txt	20	3	CP Multiple	Portfolio Search	0.1043922	0.068608	170	460	1240	1982	OPTIMAL	20	800	–	–
airland5.txt	20	3	CP Multiple	LP Search	0.1225771	0.068667	170	460	1240	1699	OPTIMAL	10	800	–	–
airland5.txt	20	4	CP Multiple	Automatic Search	0.0421937	0.068667	0	460	1240	1521	OPTIMAL	0	762	–	–
airland5.txt	20	4	CP Multiple	Fixed Search	0.0419258	0.068667	0	460	1240	708	OPTIMAL	0	760	–	–
airland5.txt	20	4	CP Multiple	Portfolio Search	0.0492757	0.068667	0	460	1240	1772	OPTIMAL	14	800	–	–
airland5.txt	20	4	CP Multiple	LP Search	0.0420881	0.068667	0	460	1240	1715	OPTIMAL	11	800	–	–
airland5.txt	20	1	Hybrid	Automatic Search	2.3587005	0.3119202	3100	841	2111	570	true	0	191	1	1
airland5.txt	20	1	Hybrid	Fixed Search	2.8876312	0.4315758	3100	841	2111	480	true	0	190	1	1
airland5.txt	20	1	Hybrid	Portfolio Search	1.6384079	0.4444351	3100	841	2111	596	true	0	216	1	1
airland5.txt	20	1	Hybrid	LP Search	2.0014477	0.4677887	3100	841	2111	284	true	0	190	1	1
airland5.txt	20	2	Hybrid	Automatic Search	0.1226707	0.3120232	650	841	2111	1016	true	0	590	1	1
airland5.txt	20	2	Hybrid	Fixed Search	0.1178794	0.4413414	650	841	2111	1428	true	0	628	1	1
airland5.txt	20	2	Hybrid	Portfolio Search	0.1313779	0.4444351	650	841	2111	0	true	0	590	1	1
airland5.txt	20	2	Hybrid	LP Search	0.1201427	0.4677887	650	841	2111	0	true	0	0	1	1
airland5.txt	20	3	Hybrid	Automatic Search	0.1723657	0.3124962	170	841	2111	1088	true	0	760	1	1
airland5.txt	20	3	Hybrid	Fixed Search	0.1583283	0.4428673	170	841	2111	1422	true	0	760	1	1
airland5.txt	20	3	Hybrid	Portfolio Search	0.1638112	0.4546814	170	841	2111	1204	true	0	760	1	1
airland5.txt	20	3	Hybrid	LP Search	0.151552	0.467617	170	841	2111	1520	true	0	760	1	1
airland5.txt	20	4	Hybrid	Automatic Search	0.0874639	0.3181419	0	841	2111	1697	true	2	760	1	1
airland5.txt	20	4	Hybrid	Fixed Search	0.0888536	0.4365883	0	841	2111	1536	true	2	760	1	1
airland5.txt	20	4	Hybrid	Portfolio Search	0.0913553	0.4577332	0	841	2111	1804	true	2	798	1	1
airland5.txt	20	4	Hybrid	LP Search	0.1022973	0.467617	0	841	2111	160	true	0	760	1	1

Fig. 18. Results for Dataset 5

Interactive Metrics Table

Filter by Dataset: airland6.txt
Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland6.txt	30	1	MIP Single	–	0.004724979400634766	0.000008	24442	960	1166	0	–	–	–	–	–
airland6.txt	30	1	MIP Multiple	–	0.0073032379150390625	0.000008	24442	1860	2182	0	–	–	–	–	–
airland6.txt	30	2	MIP Multiple	–	0.3491098880767822	0.024437	554	1890	2617	10	–	–	–	–	–
airland6.txt	30	3	MIP Multiple	–	0.33121252059936523	0.013205	0	1920	3052	64	–	–	–	–	–
airland6.txt	30	1	CP Single	Automatic Search	0.0040851	0.022396	24442	525	264	0	OPTIMAL	0	0	–	–
airland6.txt	30	1	CP Single	Fixed Search	0.0040049	0.022396	24442	525	264	0	OPTIMAL	0	0	–	–
airland6.txt	30	1	CP Single	Portfolio Search	0.0035189	0.022396	24442	525	264	0	OPTIMAL	0	0	–	–
airland6.txt	30	1	CP Single	LP Search	0.0032444	0.022396	24442	525	264	0	OPTIMAL	0	0	–	–
airland6.txt	30	1	CP Multiple	Automatic Search	0.0038115	0.084497	24442	990	1308	0	OPTIMAL	0	0	–	–
airland6.txt	30	1	CP Multiple	Fixed Search	0.0039435	0.084497	24442	990	1308	0	OPTIMAL	0	0	–	–
airland6.txt	30	1	CP Multiple	Portfolio Search	0.004303	0.084497	24442	990	1308	0	OPTIMAL	0	0	–	–
airland6.txt	30	1	CP Multiple	LP Search	0.005722	0.084497	24442	990	1308	0	OPTIMAL	0	0	–	–
airland6.txt	30	2	CP Multiple	Automatic Search	0.0314071	0.084497	554	990	1308	0	OPTIMAL	0	0	–	–
airland6.txt	30	2	CP Multiple	Fixed Search	0.043482	0.084497	554	990	1308	504	OPTIMAL	6	201	–	–
airland6.txt	30	2	CP Multiple	Portfolio Search	0.0564822	0.084497	554	990	1308	128	OPTIMAL	0	174	–	–
airland6.txt	30	2	CP Multiple	LP Search	0.0478969	0.084497	554	990	1308	0	OPTIMAL	0	174	–	–
airland6.txt	30	3	CP Multiple	Automatic Search	0.0679889	0.084497	0	990	1308	1300	OPTIMAL	0	1363	–	–
airland6.txt	30	3	CP Multiple	Fixed Search	0.0801442	0.084497	0	990	1308	2452	OPTIMAL	0	1363	–	–
airland6.txt	30	3	CP Multiple	Portfolio Search	0.1076251	0.084497	0	990	1308	136	OPTIMAL	0	1363	–	–
airland6.txt	30	3	CP Multiple	LP Search	0.0516242	0.084438	0	990	1308	1824	OPTIMAL	0	1363	–	–
airland6.txt	30	1	Hybrid	Automatic Search	0.0563879	0.3181419	24442	1512	3043	0	true	0	0	1	1
airland6.txt	30	1	Hybrid	Fixed Search	0.0586197	0.4365883	24442	1512	3043	0	true	0	0	1	1
airland6.txt	30	1	Hybrid	Portfolio Search	0.0543573	0.4577332	24442	1512	3043	0	true	0	0	1	1
airland6.txt	30	1	Hybrid	LP Search	0.0474427	0.4525528	24442	1512	3043	0	true	0	0	1	1
airland6.txt	30	2	Hybrid	Automatic Search	0.0924444	0.2930717	554	1512	3043	0	true	0	0	1	1
airland6.txt	30	2	Hybrid	Fixed Search	0.139452	0.4140129	554	1512	3043	0	true	0	0	1	1
airland6.txt	30	2	Hybrid	Portfolio Search	0.1089051	0.4184494	554	1512	3043	0	true	0	0	1	1
airland6.txt	30	2	Hybrid	LP Search	0.1140385	0.4525528	554	1512	3043	124	true	0	232	1	1
airland6.txt	30	3	Hybrid	Automatic Search	0.1468871	0.3241501	0	1512	3043	2789	true	0	1365	1	1
airland6.txt	30	3	Hybrid	Fixed Search	0.1455564	0.4359207	0	1512	3043	1392	true	0	1363	1	1
airland6.txt	30	3	Hybrid	Portfolio Search	0.1697085	0.4594955	0	1512	3043	1386	true	0	1363	1	1
airland6.txt	30	3	Hybrid	LP Search	0.1546843	0.4709244	0	1512	3043	2768	true	0	1363	1	1

Fig. 19. Results for Dataset 6

Interactive Metrics Table

Filter by Dataset: airland7.txt
Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland7.txt	44	1	MIP Single	–	0.5219495296478271	0.00514	1550	2024	2304	1575	–	–	–	–	–
airland7.txt	44	1	MIP Multiple	–	0.40361475944519043	0.011326	1550	3960	4374	1147	–	–	–	–	–
airland7.txt	44	2	MIP Multiple	–	0.023020505905151367	0.000008	0	4004	5320	1	–	–	–	–	–
airland7.txt	44	1	CP Single	Automatic Search	0.1260991	0.038014	1550	1078	347	0	OPTIMAL	0	22	–	–
airland7.txt	44	1	CP Single	Fixed Search	0.2619868	0.038014	1550	1078	347	0	OPTIMAL	0	22	–	–
airland7.txt	44	1	CP Single	Portfolio Search	0.219096	0.038014	1550	1078	347	140	OPTIMAL	0	118	–	–
airland7.txt	44	1	CP Single	LP Search	0.1855075	0.038014	1550	1078	347	0	OPTIMAL	0	22	–	–
airland7.txt	44	1	CP Multiple	Automatic Search	0.0474005	0.164873	1550	2068	2454	0	OPTIMAL	0	0	–	–
airland7.txt	44	1	CP Multiple	Fixed Search	0.0815897	0.164873	1550	2068	2454	0	OPTIMAL	0	0	–	–
airland7.txt	44	1	CP Multiple	Portfolio Search	0.0614254	0.164873	1550	2068	2454	30	OPTIMAL	0	22	–	–
airland7.txt	44	1	CP Multiple	LP Search	0.0730738	0.164873	1550	2068	2454	28	OPTIMAL	0	22	–	–
airland7.txt	44	2	CP Multiple	Automatic Search	0.0308988	0.164873	0	2068	2454	760	OPTIMAL	4	303	–	–
airland7.txt	44	2	CP Multiple	Fixed Search	0.0315801	0.164873	0	2068	2454	760	OPTIMAL	4	303	–	–
airland7.txt	44	2	CP Multiple	Portfolio Search	0.0434562	0.164873	0	2068	2454	760	OPTIMAL	4	303	–	–
airland7.txt	44	2	CP Multiple	LP Search	0.032196	0.164873	0	2068	2454	789	OPTIMAL	5	346	–	–
airland7.txt	44	1	Hybrid	Automatic Search	0.3944802	0.3272018	1550	3164	6295	436	true	101	76	2	2
airland7.txt	44	1	Hybrid	Fixed Search	0.5849545	0.4392776	1550	3164	6295	656	true	162	206	2	2
airland7.txt	44	1	Hybrid	Portfolio Search	0.4373405	0.4594955	1550	3164	6295	807	true	228	247	2	2
airland7.txt	44	1	Hybrid	LP Search	0.493849	0.4709244	1550	3164	6295	494	true	101	104	2	2
airland7.txt	44	2	Hybrid	Automatic Search	0.1479256	0.3134155	0	3119	6204	0	true	0	284	1	1
airland7.txt	44	2	Hybrid	Fixed Search	0.1440406	0.4285889	0	3119	6204	950	true	40	370	1	1
airland7.txt	44	2	Hybrid	Portfolio Search	0.1430299	0.4337349	0	3119	6204	0	true	0	284	1	1
airland7.txt	44	2	Hybrid	LP Search	0.1381931	0.4547768	0	3119	6204	568	true	0	284	1	1

Fig. 20. Results for Dataset 7

Interactive Metrics Table

Filter by Dataset: airland8.txt

Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland8.txt	50	1	MIP Single	–	3.7606468200683594	0.02134	1950	2600	4017	13	–	–	–	–	–
airland8.txt	50	1	MIP Multiple	–	3.8168764114379883	0.00505	1950	5100	8941	31	–	–	–	–	–
airland8.txt	50	2	MIP Multiple	–	12.653696775436401	0.020001	135	5150	10166	5992	–	–	–	–	–
airland8.txt	50	3	MIP Multiple	–	3.160536527633667	0.015862	0	5200	11391	85	–	–	–	–	–
airland8.txt	50	1	CP Single	Automatic Search	2.0070406	0.138519	1950	1375	2629	2468	OPTIMAL	0	1261	–	–
airland8.txt	50	1	CP Single	Fixed Search	2.3184092	0.138519	1950	1375	2629	2479	OPTIMAL	0	1272	–	–
airland8.txt	50	1	CP Single	Portfolio Search	2.227225	0.138519	1950	1375	2629	2234	OPTIMAL	0	1207	–	–
airland8.txt	50	1	CP Single	LP Search	1.5462302	0.138519	1950	1375	2629	2242	OPTIMAL	0	1207	–	–
airland8.txt	50	1	CP Multiple	Automatic Search	2.0591768	0.419723	1950	2650	7558	2228	OPTIMAL	0	1207	–	–
airland8.txt	50	1	CP Multiple	Fixed Search	1.5688041	0.419723	1950	2650	7558	2466	OPTIMAL	0	1259	–	–
airland8.txt	50	1	CP Multiple	Portfolio Search	1.4475761	0.419723	1950	2650	7558	1920	OPTIMAL	0	1207	–	–
airland8.txt	50	1	CP Multiple	LP Search	1.3384608	0.419723	1950	2650	7558	2473	OPTIMAL	0	1266	–	–
airland8.txt	50	2	CP Multiple	Automatic Search	0.2605801	0.419723	135	2650	7558	0	OPTIMAL	0	2479	–	–
airland8.txt	50	2	CP Multiple	Fixed Search	0.217771	0.419723	135	2650	7558	5607	OPTIMAL	1	2579	–	–
airland8.txt	50	2	CP Multiple	Portfolio Search	0.3117971	0.419723	135	2650	7558	7793	OPTIMAL	17	2579	–	–
airland8.txt	50	2	CP Multiple	LP Search	0.2249815	0.419723	135	2650	7558	7936	OPTIMAL	32	2639	–	–
airland8.txt	50	3	CP Multiple	Automatic Search	0.3967533	0.419723	0	2650	7558	10758	OPTIMAL	15	4987	–	–
airland8.txt	50	3	CP Multiple	Fixed Search	0.4496872	0.419723	0	2650	7558	10201	OPTIMAL	10	4987	–	–
airland8.txt	50	3	CP Multiple	Portfolio Search	0.4517218	0.419723	0	2650	7558	11077	OPTIMAL	20	4987	–	–
airland8.txt	50	3	CP Multiple	LP Search	0.439489	0.419723	0	2650	7558	10201	OPTIMAL	10	4987	–	–
airland8.txt	50	1	Hybrid	Automatic Search	1.6048102	0.3749123	1950	5093	13481	2806	true	0	1207	1	1
airland8.txt	50	1	Hybrid	Fixed Search	1.6551819	0.4816589	1950	5093	13481	2076	true	0	1207	1	1
airland8.txt	50	1	Hybrid	Portfolio Search	1.5673604	0.4960365	1950	5093	13481	2832	true	0	1207	1	1
airland8.txt	50	1	Hybrid	LP Search	1.754982	0.5080719	1950	5093	13481	2228	true	0	1207	1	1
airland8.txt	50	2	Hybrid	Automatic Search	0.6242592	0.4630051	135	5093	13481	7522	true	0	3691	1	1
airland8.txt	50	2	Hybrid	Fixed Search	0.5330243	0.5327415	135	5093	13481	5988	true	0	3691	1	1
airland8.txt	50	2	Hybrid	Portfolio Search	0.5646865	0.5422783	135	5093	13481	7504	true	0	3691	1	1
airland8.txt	50	2	Hybrid	LP Search	0.5558338	0.5449829	135	5093	13481	8077	true	0	3785	1	1
airland8.txt	50	3	Hybrid	Automatic Search	0.8175244	0.515667	0	5093	13481	9824	true	0	4887	1	1
airland8.txt	50	3	Hybrid	Fixed Search	0.7418878	0.5362511	0	5093	13481	10733	true	0	4981	1	1
airland8.txt	50	3	Hybrid	Portfolio Search	0.7443712	0.5528755	0	5093	13481	8300	true	0	4887	1	1
airland8.txt	50	3	Hybrid	LP Search	0.7992694	0.5710144	0	5093	13481	9870	true	0	4887	1	1

Fig. 21. Results for Dataset 8

Interactive Metrics Table

Filter by Dataset: airland9.txt

Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland9.txt	100	3	MIP Multiple	–	255.51801443099976	0.004937	76	20400	34643	80739	–	–	–	–	–
airland9.txt	100	4	MIP Multiple	–	76.30916404724121	0.004962	0	20500	39593	2315	–	–	–	–	–
airland9.txt	100	3	CP Multiple	Automatic Search	6.311469	1.009518	76	10300	15962	33634	OPTIMAL	10	16414	–	–
airland9.txt	100	3	CP Multiple	Fixed Search	7.1343376	1.008622	76	10300	15962	32428	OPTIMAL	0	16214	–	–
airland9.txt	100	3	CP Multiple	Portfolio Search	7.5773316	1.008622	76	10300	15962	35151	OPTIMAL	31	16414	–	–
airland9.txt	100	3	CP Multiple	LP Search	6.0204007	1.008622	76	10300	15962	34672	OPTIMAL	26	16414	–	–
airland9.txt	100	4	CP Multiple	Automatic Search	3.0474928	1.008622	0	10300	15962	35913	OPTIMAL	29	16414	–	–
airland9.txt	100	4	CP Multiple	Fixed Search	2.9552002	1.008622	0	10300	15962	34564	OPTIMAL	20	16414	–	–
airland9.txt	100	4	CP Multiple	Portfolio Search	3.849262	1.008622	0	10300	15962	34733	OPTIMAL	20	16414	–	–
airland9.txt	100	4	CP Multiple	LP Search	1.8586243	1.008622	0	10300	15962	36293	OPTIMAL	29	16414	–	–
airland9.txt	100	3	Hybrid	Automatic Search	41.3386903	1.7210541	76	17074	37651	159482	false	0	14156	5	5
airland9.txt	100	3	Hybrid	Fixed Search	71.4933662	7.1852989	76	17074	37651	161234	false	0	15904	5	5
airland9.txt	100	3	Hybrid	Portfolio Search	83.0878913	6.7347832	76	17074	37651	150327	false	0	12795	5	5
airland9.txt	100	3	Hybrid	LP Search	73.6607656	7.4889488	76	17074	37651	149081	false	0	14478	5	5
airland9.txt	100	4	Hybrid	Automatic Search	3.1650302	1.7244797	0	16670	36839	33594	true	21	16214	1	1
airland9.txt	100	4	Hybrid	Fixed Search	2.6287768	1.6204453	0	16670	36839	58936	true	50	16214	1	1
airland9.txt	100	4	Hybrid	Portfolio Search	2.9481916	1.4036865	0	16670	36839	49294	true	50	16214	1	1
airland9.txt	100	4	Hybrid	LP Search	5.7944515	1.3645058	0	16670	36839	31836	true	0	16214	1	1

Fig. 22. Results for Dataset 9

Interactive Metrics Table

Filter by Dataset: airland10.txt

Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland10.txt	150	4	MIP Multiple	–	908.1463239192963	0.005636	34	45750	85772	354358	–	–	–	–	–
airland10.txt	150	5	MIP Multiple	–	164.40255427360535	0.005017	0	45900	96947	6002	–	–	–	–	–
airland10.txt	150	4	CP Multiple	Automatic Search	11.227598	2.095493	34	22950	31616	72583	OPTIMAL	11	35924	–	–
airland10.txt	150	4	CP Multiple	Fixed Search	9.0984345	2.095493	34	22950	31616	75427	OPTIMAL	26	35924	–	–
airland10.txt	150	4	CP Multiple	Portfolio Search	10.6056964	2.096487	34	22950	31616	125029	OPTIMAL	50	35924	–	–
airland10.txt	150	4	CP Multiple	LP Search	11.1173238	2.095493	34	22950	31616	77409	OPTIMAL	41	35924	–	–
airland10.txt	150	5	CP Multiple	Automatic Search	9.5866472	2.095493	0	22950	31616	72113	OPTIMAL	10	35924	–	–
airland10.txt	150	5	CP Multiple	Fixed Search	6.9116717	2.095493	0	22950	31616	72931	OPTIMAL	21	35924	–	–
airland10.txt	150	5	CP Multiple	Portfolio Search	10.1844265	2.095493	0	22950	31616	75836	OPTIMAL	32	35924	–	–
airland10.txt	150	5	CP Multiple	LP Search	9.8924277	2.095493	0	22950	31616	65876	OPTIMAL	0	35624	–	–
airland10.txt	150	4	Hybrid	Automatic Search	103.9049609	2.7871895	37	36899	79178	353509	false	0	35526	5	5
airland10.txt	150	4	Hybrid	Fixed Search	161.0636306	2.9844093	37	36899	79178	357225	false	10	36112	5	5
airland10.txt	150	4	Hybrid	Portfolio Search	175.4606972	3.5984116	37	36899	79178	455059	false	54	36171	5	5
airland10.txt	150	4	Hybrid	LP Search	156.1400001	3.4976082	37	36899	79178	347423	false	1	35791	5	5
airland10.txt	150	5	Hybrid	Automatic Search	7.1677451	2.6535835	0	36295	77966	110652	true	10	35624	1	1
airland10.txt	150	5	Hybrid	Fixed Search	7.5651839	2.5422516	0	36295	77966	119440	true	50	35624	1	1
airland10.txt	150	5	Hybrid	Portfolio Search	7.8214903	3.5442352	0	36295	77966	102576	true	50	35624	1	1
airland10.txt	150	5	Hybrid	LP Search	7.9073877	3.4343491	0	36295	77966	63922	true	0	35624	1	1

Fig. 23. Results for Dataset 10

Interactive Metrics Table

Filter by Dataset: airland11.txt

Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland11.txt	200	5	MIP Multiple	–	212.1592652797699	0.004964	0	81200	169472	2001	–	–	–	–	–
airland11.txt	200	5	CP Multiple	Automatic Search	23.0743875	3.659226	0	40600	52456	234808	OPTIMAL	50	62946	–	–
airland11.txt	200	5	CP Multiple	Fixed Search	37.0735708	3.659226	0	40600	52456	216061	OPTIMAL	50	62946	–	–
airland11.txt	200	5	CP Multiple	Portfolio Search	29.0342812	3.659226	0	40600	52456	127018	OPTIMAL	18	62946	–	–
airland11.txt	200	5	CP Multiple	LP Search	31.7378089	3.659226	0	40600	52456	211383	OPTIMAL	20	62946	–	–
airland11.txt	200	5	Hybrid	Automatic Search	67.0270061	3.1603127	0	63485	134381	129670	true	10	62546	1	1
airland11.txt	200	5	Hybrid	Fixed Search	78.2393634	3.3969841	0	63485	134381	120280	true	0	62546	1	1
airland11.txt	200	5	Hybrid	Portfolio Search	64.1553619	3.7431221	0	63485	134381	177219	true	50	62546	1	1
airland11.txt	200	5	Hybrid	LP Search	112.8958056	3.6700974	0	63485	134381	123988	true	0	62546	1	1

Fig. 24. Results for Dataset 11

Interactive Metrics Table

Filter by Dataset: airland12.txt

Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland12.txt	250	6	MIP Multiple	–	49.43548846244812	0.005221	0	126750	293710	403	–	–	–	–	–
airland12.txt	250	6	CP Multiple	Automatic Search	53.6752141	5.596086	0	63250	79062	412913	OPTIMAL	50	97679	–	–
airland12.txt	250	6	CP Multiple	Fixed Search	62.6268179	5.596086	0	63250	79062	429021	OPTIMAL	20	97679	–	–
airland12.txt	250	6	CP Multiple	Portfolio Search	52.5705453	5.596086	0	63250	79062	378911	OPTIMAL	20	97679	–	–
airland12.txt	250	6	CP Multiple	LP Search	24.6668342	5.596086	0	63250	79062	313656	OPTIMAL	50	97679	–	–
airland12.txt	250	6	Hybrid	Automatic Search	109.1695898	4.5883789	0	98353	206713	265975	true	50	97179	1	1
airland12.txt	250	6	Hybrid	Fixed Search	111.863862	4.8519936	0	98353	206713	201312	true	46	97179	1	1
airland12.txt	250	6	Hybrid	Portfolio Search	104.013561	5.3122559	0	98353	206713	272067	true	50	97179	1	1
airland12.txt	250	6	Hybrid	LP Search	146.9954073	5.1664047	0	98353	206713	191360	true	0	97179	1	1

Fig. 25. Results for Dataset 12

Interactive Metrics Table

Filter by Dataset: airland13.txt

Filter by Solver: Select solver(s)

Dataset	#Planes	#Runways	Solver	Strategy	Time (s)	Memory (MB)	Penalty / Obj	#Variables	#Constraints	#Branches	Converged	Num Conflicts	Num Booleans	Num Iterations	MIP Num Calls
airland13.txt	500	7	MIP Multiple	–	225.1651475429535	0.004897	0	504000	1276434	3734	–	–	–	–	–
airland13.txt	500	7	CP Multiple	Automatic Search	358.7680958	21.019262	0	251500	285486	377112	OPTIMAL	17	384150	–	–
airland13.txt	500	7	CP Multiple	Fixed Search	733.2623064	21.019262	0	251500	285486	372653	OPTIMAL	13	383455	–	–
airland13.txt	500	7	CP Multiple	Portfolio Search	1002.1035842	21.019262	0	251500	285486	377112	OPTIMAL	17	384150	–	–
airland13.txt	500	7	CP Multiple	LP Search	326.7198216	21.019262	0	251500	285486	706564	OPTIMAL	50	383455	–	–
airland13.txt	500	7	Hybrid	Automatic Search	1109.3105214	8.1080093	0	384803	791400	1045425	true	50	382456	1	1
airland13.txt	500	7	Hybrid	Fixed Search	806.4051225	7.8203888	0	384803	791400	1153928	true	62	382455	1	1
airland13.txt	500	7	Hybrid	Portfolio Search	956.2799654	8.2414474	0	384803	791400	916344	true	20	382455	1	1
airland13.txt	500	7	Hybrid	LP Search	1158.1066854	7.6774712	0	384803	791400	532194	true	10	382455	1	1

Fig. 26. Results for Dataset 13